

Engineering Mathematics

Previous Year Questions

DIFFERENTIAL EQUATIONS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Differential Equations (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:



 Easy  Medium  Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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Contents

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UNIT

I

Differential Equations

Topics

1. Order, Degree, Formation and Solution of a Differential Equation
2. Homogeneous, Exact and Linear Differential Equations
3. Non-Exact Differential Equations
4. Non-Homogeneous Differential Equations
5. Solution of Higher Order ODEs
6. Cauchy-Euler's ODE
7. Partial Differential Equations

Note to Reader

Differential Equations are present in branches with paper codes AE, AG, BT, BM, CE, CH, EC, EE, ES, IN, MA, ME, MN, NM, PE, PI, XE according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Branches: EC,EE,IN,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 38 **MCQ:** 32 **MSQ:** 1 **NAT:** 5
1M: 22 **2M:** 16 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2020)

EE — Electrical Engineering

Questions: 18 **MCQ:** 12 **MSQ:** 1 **NAT:** 5
1M: 9 **2M:** 9 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–3M(2026)

IN — Instrumentation Engineering

Questions: 17 **MCQ:** 15 **MSQ:** 1 **NAT:** 1
1M: 5 **2M:** 12 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–3M(2026)

ME — Mechanical Engineering

Questions: 56 **MCQ:** 48 **MSQ:** 0 **NAT:** 8
1M: 23 **2M:** 31 **Desc:** 2 **Avg Weightage** ~2M **Range:** 1M(2016)–3M(2026)

CE — Civil Engineering

Questions: 64 **MCQ:** 51 **MSQ:** 3 **NAT:** 10
1M: 27 **2M:** 36 **Desc:** 1 **Avg Weightage** ~3M **Range:** 1M(2022)–5M(2010)

CH — Chemical Engineering

Questions: 36 **MCQ:** 29 **MSQ:** 1 **NAT:** 6
1M: 11 **2M:** 25 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2023)–4M(2014)

PI — Production & Industrial Engineering

Questions: 24 **MCQ:** 20 **MSQ:** 0 **NAT:** 4
1M: 12 **2M:** 12 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2023)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q6.26.1 MCQ 2026 1M

Consider the differential equation $\dot{\vec{w}} = A\vec{w}$, with $\vec{w}(t = 0) = \begin{bmatrix}1\\1\end{bmatrix}$.

If $\vec{w}(t) = e^{t}\vec{u}_x + e^{-2t}\vec{u}_y$ be the solution to the equation where $\vec{u}_x$ and $\vec{u}_y$ are unit vectors along the positive x and y axes respectively, then which of the following options is the correct matrix representing A?

A. $\begin{bmatrix}1 & 0\\0 & -2\end{bmatrix}$ **B.** $\begin{bmatrix}-1 & 0\\0 & 2\end{bmatrix}$

C. $\begin{bmatrix}0 & -2\\1 & 0\end{bmatrix}$ **D.** $\begin{bmatrix}0 & 2\\-1 & 0\end{bmatrix}$

Q6.25.1 NAT 2025 1M

The function $y(t)$ satisfies

$$t^2y''(t) - 2ty'(t) + 2y(t) = 0,$$

where $y'(t)$ and $y''(t)$ denote the first and second derivatives of $y(t)$, respectively.

Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0, 1]$ is _____ *(rounded off to two decimal places).*

Q6.24.1 MCQ 2024 1M

The general form of the complementary function of a differential equation is given by $y(t) = (At+B)e^{-2t}$, where A and B are real constants determined by the initial condition. The corresponding differential equation is _____.

A. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = f(t)$

B. $\frac{d^2y}{dt^2} + 4y = f(t)$

C. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = f(t)$

D. $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = f(t)$

Q6.22.1 MSQ 2022 1M

Consider the following partial differential equation (PDE)

$$a\frac{\partial^2 f(x,y)}{\partial x^2} + b\frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y),$$

where a and b are distinct positive real numbers. Select the combination(s) of values of the real parameters ξ and η such that $f(x, y) = e^{(\xi x+\eta y)}$ is a solution of the given PDE.

☐ **A.** $\xi = \frac{1}{\sqrt{2a}}$, $\eta = \frac{1}{\sqrt{2b}}$

☐ **B.** $\xi = \frac{1}{\sqrt{a}}$, $\eta = 0$

☐ **C.** $\xi = 0$, $\eta = 0$

☐ **D.** $\xi = \frac{1}{\sqrt{a}}$, $\eta = \frac{1}{\sqrt{b}}$

Q6.21.1 MCQ 2021 1M

Consider the differential equation given below.

$$\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$$

The integrating factor of the differential equation is

A. $(1-x^2)^{-3/4}$ **B.** $(1-x^2)^{-1/4}$

C. $(1-x^2)^{-3/2}$ **D.** $(1-x^2)^{-1/2}$

Q6.20.1 MCQ 2020 1M

The partial derivative of the function

$$f(x, y, z) = e^{1-x\cos y} + xze^{-1/(1+y^2)}$$

with respect to x at the point $(1, 0, e)$ is

A. -1 **B.** 0 **C.** 1 **D.** $\frac{1}{e}$

Q6.20.2 MCQ 2020 1M

The general solution of $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$ is

A. $y = C_1e^{3x} + C_2e^{-3x}$ **B.** $y = (C_1 + C_2x)e^{-3x}$

C. $y = (C_1 + C_2x)e^{3x}$ **D.** $y = C_1e^{3x}$

Q6.20.3 MCQ 2020 2M

Which one of the following options contain two solutions of the differential equation $\frac{dy}{dx} = (y-1)x$?

A. $\ln|y-1| = 0.5x^2 + C$ and $y = 1$

B. $\ln|y-1| = 2x^2 + C$ and $y = 1$

C. $\ln|y-1| = 0.5x^2 + C$ and $y = -1$

D. $\ln|y-1| = 2x^2 + C$ and $y = -1$

Q6.19.1 MCQ 2019 1M

The families of curves represented by the solution of the equation

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^n$$

for $n = -1$ and $n = +1$, respectively, are

A. Parabolas and Circles

B. Circles and Hyperbolas

C. Hyperbolas and Circles

D. Hyperbolas and Parabolas

Q6.19.2 NAT 2019 2M

Consider the homogeneous ordinary differential equation

$$x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 3y = 0, \qquad x > 0$$

with $y(x)$ as a general solution. Given that $y(1) = 1$ and $y(2) = 14$. The value of $y(1.5)$, rounded off to two decimal places, is _____

Q6.18.1 MCQ 2018 2M

A curve passes through the point $(x = 1, y = 0)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2}{2y} + \frac{y}{x}$. The equation that describes the curve is

A. $\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$

B. $\frac{1}{2}\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$

C. $\ln\left(1 + \frac{y}{x}\right) = x - 1$

D. $\frac{1}{2}\ln\left(1 + \frac{y}{x}\right) = x - 1$

Q6.18.2 NAT 2018 2M

The position of a particle $y(t)$ is described by the differential equation:

$$\frac{d^2y}{dt^2} = -\frac{dy}{dt} - \frac{5y}{4}.$$

The initial conditions are $y(0) = 1$ and $\left.\frac{dy}{dt}\right|_{t=0} = 0$. The position (accurate to two decimal places) of the particle at $t = \pi$ is _____

Q6.17.1 MCQ 2017 2M

Which one of the following is the general solution of the first order differential equation

$$\frac{dy}{dx} = (x + y - 1)^2,$$

where x, y are real?

A. $y = 1 + x + \tan^{-1}(x + c)$, where c is a constant

B. $y = 1 + x + \tan(x + c)$, where c is a constant

C. $y = 1 - x + \tan^{-1}(x + c)$, where c is a constant

D. $y = 1 - x + \tan(x + c)$, where c is a constant

Q6.17.2 MCQ 2017 1M

The general solution of the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 5y = 0$$

in terms of arbitrary constants K_1 and K_2 is

A. $K_1e^{(-1+\sqrt{6})x} + K_2e^{(-1-\sqrt{6})x}$

B. $K_1e^{(-1+\sqrt{8})x} + K_2e^{(-1-\sqrt{8})x}$

C. $K_1e^{(-2+\sqrt{6})x} + K_2e^{(-2-\sqrt{6})x}$

D. $K_1e^{(-2+\sqrt{8})x} + K_2e^{(-2-\sqrt{8})x}$

Q6.16.1 MCQ 2016 2M

The particular solution of the Initial value problem given below is

$$\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0$$

with $y(0) = 3$ and $\left.\frac{dy}{dx}\right|_{x=0} = -36$

A. $(3 - 18x)\,e^{-6x}$ **B.** $(3 + 25x)\,e^{-6x}$

C. $(3 + 20x)\,e^{-6x}$ **D.** $(3 - 12x)\,e^{-6x}$

Q6.15.1 MCQ 2015 2M

The solution of the differential equation $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$ with $y(0) = y'(0) = 1$ is

A. $(2 - t)\,e^t$ **B.** $(1 + 2t)\,e^{-t}$

C. $(2 + t)\,e^{-t}$ **D.** $(1 - 2t)\,e^t$

Q6.15.2 MCQ 2015 1M

The general solution of the differential equation $\frac{dy}{dx} = \frac{1+\cos 2y}{1-\cos 2x}$ is

A. $\tan y - \cot x = c$ (c is a constant)

B. $\tan x - \cot y = c$ (c is a constant)

C. $\tan y + \cot x = c$ (c is a constant)

D. $\tan x + \cot y = c$ (c is a constant)

Q6.15.3 MCQ 2015 2M

Consider the differential equation $\frac{dx}{dt} = 10 - 0.2x$ with initial condition $x(0) = 1$. The response $x(t)$ for $t > 0$ is

A. $2 - e^{-0.2t}$ **B.** $2 - e^{0.2t}$

C. $50 - 49e^{-0.2t}$ **D.** $50 - 49e^{0.2t}$

Q6.15.4 NAT 2015 2M

Consider the differential equation

$$\frac{d^2x(t)}{dt^2} + 3\frac{dx(t)}{dt} + 2x(t) = 0.$$

with $y(x)$ Given $x(0) = 20$ and $x(1) = 10/e$, where $e = 2.718$, the value of $x(2)$ is _____

Q6.14.1 MCQ 2014 1M

If the characteristic equation of the differential equation

$$\frac{d^2y}{dx^2} + 2\alpha\frac{dy}{dx} + y = 0$$

has two equal roots, then the values of α are

A. ± 1 **B.** $0, 0$ **C.** $\pm j$ **D.** $\pm 1/2$

Q6.14.2 MCQ 2014 1M

Which ONE of the following is a linear non-homogeneous differential equation, where x and y are the independent and dependent variables respectively?

A. $\frac{dy}{dx} + xy = e^{-x}$ **B.** $\frac{dy}{dx} + xy = 0$

C. $\frac{dy}{dx} + xy = e^{-y}$ **D.** $\frac{dy}{dx} + e^{-y} = 0$

Q6.14.3 MCQ 2014 1M

If $z = xy\ln(xy)$, then

A. $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0$ **B.** $y\frac{\partial z}{\partial x} = x\frac{\partial z}{\partial y}$

C. $x\frac{\partial z}{\partial x} = y\frac{\partial z}{\partial y}$ **D.** $y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = 0$

Q6.14.4 MCQ 2014 1M

If a and b are constants, the most general solution of the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 0 \text{ is}$$

A. ae^{-t} **B.** $ae^{-t} + bte^{-t}$

C. $ae^{t} + bte^{-t}$ **D.** ae^{-2t}

Q6.14.5 NAT 2014 2M

With initial values $y(0) = y'(0) = 1$, the solution of the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$$

at $x = 1$ is _____.

Q6.13.1 MCQ 2013 2M

A system is described by a linear, constant coefficient, ordinary, first order differential equation has an exact solution given by $y(t)$ for $t > 0$, when the forcing function is $x(t)$ and the initial condition is $y(0)$. If one wishes to modify the system so that the solution becomes $-2y(t)$ for $t > 0$, we need to

A. change the initial condition to $-y(0)$ and the forcing function to $2x(t)$

B. change the initial condition to $2y(0)$ and the forcing function to $-x(t)$

C. change the initial condition to $j\sqrt{2}\,y(0)$ and the forcing function to $j\sqrt{2}\,x(t)$

D. change the initial condition to $-2y(0)$ and the forcing function to $-2x(t)$

Q6.12.1 MCQ 2012 1M

With initial condition $x(1) = 0.5$, the solution of the differential equation, $t\frac{dx}{dt} + x = t$ is

A. $x = t - \frac{1}{2}$ **B.** $x = t^2 - \frac{1}{2}$

C. $x = \frac{t^2}{2}$ **D.** $x = \frac{t}{2}$

Q6.12.2 MCQ 2012 2M

Consider the differential equation

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = \delta(t) \text{ with } y(t)|_{t=0^-} = -2$$

and $\left.\frac{dy}{dt}\right|_{t=0^-} = 0$. The numerical value of $\left.\frac{dy}{dt}\right|_{t=0^+}$ is

A. -2 **B.** -1 **C.** 0 **D.** 1

Q6.11.1 MCQ 2011 1M

The solution of the differential equation $\frac{dy}{dx} = ky,\ y(0) = c$ is

A. $x = ce^{-ky}$ **B.** $x = ke^{cy}$

C. $y = ce^{kx}$ **D.** $y = ce^{-kx}$

Q6.10.1 MCQ 2010 1M

A function $n(x)$ satisfies the differential equation

$$\frac{d^2n(x)}{dx^2} - \frac{n(x)}{L^2} = 0$$

where L is a constant. The boundary conditions are: $n(0) = K$ and $n(\infty) = 0$. The solution of this equation is

A. $n(x) = Ke^{x/L}$ **B.** $n(x) = Ke^{-x/\sqrt{L}}$

C. $n(x) = K^2 e^{-x/L}$ **D.** $n(x) = Ke^{-x/L}$

Q6.09.1 MCQ 2009 1M

The order of the differential equation

$$\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^3 + y^4 = e^{-t} \text{ is}$$

A. 1 **B.** 2 **C.** 3 **D.** 4

Q6.09.2 MCQ 2009 2M

Match each differential equation in Group I to its family of solution curves from Group II.

	Group I		Group II
P.	$\frac{dy}{dx} = \frac{y}{x}$	1.	Circles
Q.	$\frac{dy}{dx} = -\frac{y}{x}$	2.	Straight lines
R.	$\frac{dy}{dx} = \frac{x}{y}$	3.	Hyperbolas
S.	$\frac{dy}{dx} = -\frac{x}{y}$		

A. P-2, Q-3, R-3, S-1 **B.** P-1, Q-3, R-2, S-1

C. P-2, Q-1, R-3, S-3 **D.** P-3, Q-2, R-1, S-2

Q6.08.1 MCQ 2008 1M

Which of the following is a solution of the differential equation $\frac{dx(t)}{dt} + 3x(t) = 0$ with initial condition x(0)=2?

A. $x(t) = 3e^{-t}$ **B.** $x(t) = 2e^{-3t}$

C. $-\frac{3}{2}t^2$ **D.** $3t^2$

Q6.07.1 MCQ 2007 2M

The solution of the differential equation $k^2\frac{d^2y}{dx^2} = y - y_2$ under the boundary conditions $(i)\ y = y_1$ at $x = 0$ and $(ii)\ y = y_2$ at $x = \infty$, where k, y_1 and y_2 are constants, is

A. $y = (y_1 - y_2)exp(-x/k^2) + y_2$

B. $y = (y_2 - y_1)exp(-x/k) + y_1$

C. $y = (y_1 - y_2)sinh(x/k) + y_1$

D. $y = (y_1 - y_2)exp(-x/k) + y_2$

Q6.06.1 MCQ 2006 1M

A solution for the differential equation $\dot{x}(t) + 2x(t) = \delta(t)$ with initial condition $x(0^-) = 0$ is

A. $e^{-2t}\,u(t)$ **B.** $e^{2t}\,u(t)$

C. $e^{-t}\,u(t)$ **D.** $e^{t}\,u(t)$

Q6.06.2 MCQ 2006 2M

For the differential equation $\frac{d^2y}{dx^2} + k^2y = 0$ the boundary conditions are (i) $y = 0$ for $x = 0$ and (ii) $y = 0$ for $x = a$.

The form of non-zero solution of y (where m varies over all integers) are

A. $y = \sum_m A_m \sin \frac{m\pi x}{a}$ **B.** $y = \sum_m A_m \cos \frac{m\pi x}{a}$

C. $y = \sum_m A_m x^{\frac{m\pi}{a}}$ **D.** $y = \sum_m A_m e^{\frac{-m\pi x}{a}}$

Q6.05.1 MCQ 2005 1M

The following differential equation has

$$3\left(\frac{d^2y}{dx^2}\right) + 4\left(\frac{dy}{dx}\right)^3 + y^2 + 2 = x$$

A. degree=2, order=1

B. degree=1, order=2

C. degree=4, order=3

D. degree=2, order=3

Q6.05.2 MCQ 2005 1M

A solution of the following differential equation is given by $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

A. $y = e^{2x} + e^{-3x}$ **B.** $y = e^{2x} + e^{3x}$

C. $y = e^{-2x} + e^{3x}$ **D.** $y = e^{-2x} + e^{-3x}$

Q6.94.1 MCQ 1994 2M

Match each of the items A, B and C with an appropriate item from 1, 2, 3, 4 and 5.

(A) $a_1\frac{d^2y}{dx^2} + a_2\, y\frac{dy}{dx} + a_3\, y = a_4$

(B) $a_1\frac{d^3y}{dx^3} + a_2\, y = a_3$

(C) $a_1\frac{d^2y}{dx^2} + a_2\, x\frac{dy}{dx} + a_3\, x^2y = 0$

1. Non-linear differential equation
2. Linear differential equation with constant coefficient
3. Linear homogeneous differential equation
4. Non-linear homogeneous differential equation
5. Non-linear first order differential equation

A. A–1, B–2, C–4

B. A–3, B–4, C–2

C. A–2, B–5, C–3

D. A–3, B–1, C–2

Electrical Engineering

Q6.26.1 MCQ 2026 2M

Consider the second-order differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

with initial conditions

$$y(0) = 1, \qquad \left.\frac{dy}{dx}\right|_{x=0} = 1$$

The solution is given by

A. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

B. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

C. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

D. $y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) - \sqrt{3}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

Q6.26.2 MCQ 2026 1M

Consider the following differential equation:

$$t^2\frac{d^2y}{dt^2} + 7t\frac{dy}{dt} + 8ty = 10\sin(t)$$

Which one of the following options is correct?

A. It is a linear differential equation

B. It is a nonlinear differential equation

C. It is a time-invariant differential equation

D. It is a second-order partial differential equation

Q6.25.1 NAT 2025 2M

Consider ordinary differential equations given by

$$\dot{x_1}(t) = 2x_2(t)$$

$$\dot{x_2}(t) = r(t)$$

with initial conditions $x_1(0) = 1$ and $x_2(0) = 0$.

If $r(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$, then at $t = 1$, $x_1(t) =$ _____

(round off to the nearest integer).

Q6.24.1 MSQ 2024 2M

Which of the following differential equations is/are nonlinear?

☐ **A.** $t\,x(t) + \dfrac{dx(t)}{dt} = t^2 e^t, \quad x(0) = 0$

☐ **B.** $\dfrac{1}{2}e^t + x(t)\dfrac{dx(t)}{dt} = 0, \quad x(0) = 0$

☐ **C.** $x(t)\cos t - \dfrac{dx(t)}{dt}\sin t = 1, \quad x(0) = 0$

☐ **D.** $x(t) + e^{\left(\frac{dx(t)}{dt}\right)} = 1, \quad x(0) = 0$

Q6.20.1 NAT 2020 1M

Consider the initial value problem below. The value of y at $x = \ln 2$, (rounded off to 3 decimal places) is _____.

$$\frac{dy}{dx} = 2x - y, \qquad y(0) = 1$$

Q6.19.1 MCQ 2019 1M

The partial differential equation

$$\frac{\partial^2 u}{\partial t^2} - c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0; \text{ where } c \neq 0$$

is known as

A. Heat Equation **B.** Wave Equation

C. Poisson's Equation **D.** Laplace Equation

Q6.17.1 MCQ 2017 2M

Consider the differential equation $(t^2 - 81)\dfrac{dy}{dt} + 5t\,y = \sin(t)$ with $y(1) = 2\pi$. There exists a unique solution for this differential equation when t belongs to the interval

A. $(-2, 2)$ **B.** $(-10, 10)$ **C.** $(-10, 2)$ **D.** $(0, 10)$

Q6.16.1 MCQ 2016 1M

A function $y(t)$, such that $y(0) = 1$ and $y(1) = 3e^{-t}$ is a solution of the differential equation $\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt} + y = 0$. Then $y(2)$ is

A. $5e^{-t}$ **B.** $5e^{-2}$ **C.** $7e^{-t}$ **D.** $7e^{-2}$

Q6.16.2 MCQ 2016 1M

The solution of the differential equation, for $t > 0$, $y''(t) + 2y'(t) + y(t) = 0$ with initial conditions $y(0) = 0$ and $y'(0) = 1$, is ($u(t)$ denotes the unit step function),

A. $te^{-t}u(t)$ **B.** $e^{-t} - te^{-t}u(t)$

C. $(-e^{-t} + te^{-t})u(t)$ **D.** $e^{-t}u(t)$

Q6.16.3 NAT 2016 2M

Let $y(x)$ be the solution of the differential equation $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = 0$ with initial conditions $y(0) = 0$ and $\left.\dfrac{dy}{dx}\right|_{x=0} = 1$. Then the value of $y(1)$ is _____.

Q6.15.1 NAT 2015 2M

A solution of the ordinary differential equation $\dfrac{d^2y}{dt^2} + 5\dfrac{dy}{dt} + 6y = 0$ is such that $y(0) = 2$ and $y(1) = -\dfrac{1 - 3e}{e^3}$. The value of $\dfrac{dy}{dt}(0)$ is _____.

Q6.15.2 NAT 2015 2M

A differential equation $\dfrac{di}{dt} - 0.2i = 0$ is applicable over $-10 < t < 10$. If $i(4) = 10$, then $i(-5)$ is _____.

Q6.14.1 MCQ 2014 1M

The solution for the differential equation

$$\frac{d^2x}{dt^2} = -9x,$$

with initial conditions $x(0) = 1$ and $\left.\dfrac{dx}{dt}\right|_{t=0} = 1$, is

A. $t^2 + t + 1$ **B.** $\sin 3t + \dfrac{1}{3}\cos 3t + \dfrac{2}{3}$

C. $\dfrac{1}{3}\sin 3t + \cos 3t$ **D.** $\cos 3t + t$

Q6.14.2 MCQ 2014 1M

Consider the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.

Which of the following is a solution to this differential equation for $x > 0$?

A. e^x **B.** x^2 **C.** $1/x$ **D.** $\ln x$

Q6.11.1 MCQ 2011 1M

With K as a constant, the possible solution for the first order differential equation $\frac{dy}{dx} = e^{-3x}$ is

A. $-\frac{1}{3}e^{-3x} + K$ **B.** $-\frac{1}{3}e^{3x} + K$

C. $-3e^{-3x} + K$ **D.** $-3e^{-x} + K$

Q6.10.1 MCQ 2010 2M

For the differential equation $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 0$ with initial conditions $x(0) = 1$ and $\left.\frac{dx}{dt}\right|_{t=0} = 0$, the solution is

A. $x(t) = 2e^{-6t} - e^{-2t}$ **B.** $x(t) = 2e^{-2t} - e^{-4t}$

C. $x(t) = -e^{-6t} + 2e^{-4t}$ **D.** $x(t) = e^{-2t} + 2e^{-4t}$

Q6.05.1 MCQ 2005 1M

The solution of the first order differential equation $\dot{x}(t) = -3x(t)$, $x(0) = x_0$ is:

A. $x(t) = x_0\, e^{-3t}$ **B.** $x(t) = x_0\, e^{-3}$

C. $x(t) = x_0\, e^{-t/3}$ **D.** $x(t) = x_0\, e^{-t}$

Q6.05.2 MCQ 2005 2M

For the equation $\ddot{x}(t) + 3\dot{x}(t) + 2x(t) = 5$, the solution $x(t)$ approaches the following values at $t \to \infty$

A. 0 **B.** $\frac{5}{2}$ **C.** 5 **D.** 10

Instrumentation Engineering

Q6.26.1 MCQ 2026 1M

Among the following, the differential equation(s) not representative of a linear dynamical system is/are _____.

(i) $\frac{d^2x}{dt^2} + 2t\frac{dx}{dt} + x(t) = u(t)$

(ii) $\left(\frac{dx}{dt}\right)^2 + 2\frac{dx}{dt} + x(t) = u(t)$

(iii) $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x(t) = u(t)$

A. (i) and (ii) only **B.** (i) only

C. (ii) only **D.** (ii) and (iii) only

Q6.26.2 MSQ 2026 2M

Given a differential equation

$$\frac{d^2x}{dt^2} + x = 0 \text{ with } x(0) \neq 0.$$

Which of the following statements is/are true?

☐ **A.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = 0$ for all $t \geq 0$

☐ **B.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = c$ for all $t \geq 0$, for some real constant $c > 0$

☐ **C.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = e^{jt}$ for all $t \geq 0$

☐ **D.** $|x(t)|^2 + \left|\frac{dx(t)}{dt}\right|^2 = \sin t$ for all $t \geq 0$

Q6.25.1 MCQ 2025 1M

The solution of the differential equation $\frac{dy}{dx} = 9\frac{x}{y}$ represents

A. a hyperbola **B.** a parabola

C. an ellipse **D.** a circle

Q6.24.1 NAT 2024 2M

The solution of an ordinary differential equation $y''' + 3y'' + 3y' + y = 30\,e^{-t}$ is

$$y(t) = (c_0 + c_1 t - c_2 t^2 + c_3 t^3)e^{-t}$$

Given $y(0) = 3$, $y'(0) = -3$ and $y''(0) = -47$, the value of $(c_0 + c_1 + c_2 + c_3)$ is _____ **(rounded off to nearest integer).**

Note: $y''' = d^3y/dt^3$, $y'' = d^2y/dt^2$, $y' = dy/dt$ and c_0, c_1, c_2, c_3 are constants.

Q6.23.1 MCQ 2023 1M

The solution $x(t)$, $t \geq 0$, to the differential equation $\ddot{x} = -k\dot{x}$, $k > 0$ with initial conditions $x(0) = 1$ and $\dot{x}(0) = 0$ is

A. $x(t) = 2e^{-kt} + 2kt - 1$

B. $x(t) = 2e^{-kt} - 1$

C. $x(t) = 1$

D. $x(t) = 2e^{-kt} - kt - 1$

Q6.22.1 MCQ 2022 2M

Consider the differential equation

$$\frac{dy}{dx} + y \ln(y) = 0$$

If $y(0) = e$, then $y(1)$ is _____

A. e^e **B.** e^{-e} **C.** $e^{(1/e)}$ **D.** $e^{(-1/e)}$

Q6.20.1 MCQ 2020 2M

Consider the differential equation $\frac{dx}{dt} = \sin(x)$, with the initial condition $x(0) = 0$. The solution to this ordinary differential equation is _____

A. $x(t) = 0$ **B.** $x(t) = \sin(t)$

C. $x(t) = \cos(t)$ **D.** $x(t) = \sin(t) - cos(t)$

Q6.18.1 MCQ 2018 2M

Consider the following equation

$$\frac{\partial V(x,y)}{\partial x} = px^2 + y^2 + 2xy$$

$$\frac{\partial V(x,y)}{\partial y} = x^2 + qy^2 + 2xy$$

where p and q are constants. $V(x, y)$ that satisfies the above equations is

A. $p\frac{x^3}{3} + q\frac{y^3}{3} + 2xy + 6$

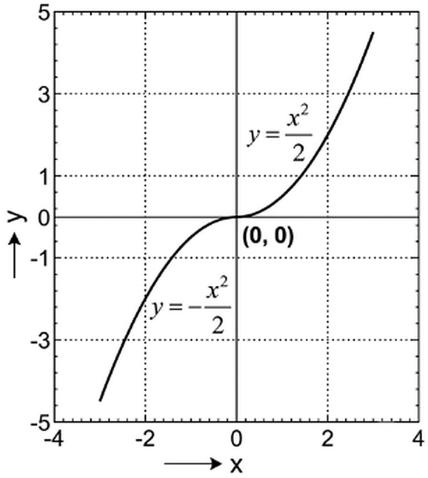
B. $p\frac{x^3}{3} + q\frac{y^3}{3} + 5$

C. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2 + xy$

D. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2$

Q6.14.1 MCQ 2014 1M

The figure shows the plot of y as a function of x



The function shown is the solution of the differential equation (assuming all initial conditions to be zero) is:

A. $\frac{d^2y}{dx^2} = 1$ **B.** $\frac{dy}{dx} = x$

C. $\frac{dy}{dx} = -x$ **D.** $\frac{dy}{dx} = |x|$

Q6.13.1 MCQ 2013 1M

The type of the partial differential equation $\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$ is

A. Parabolic **B.** Elliptic

C. Hyperbolic **D.** Nonlinear

Q6.13.2 MCQ 2013 2M

The maximum value of the solution $y(t)$ of the differential equation $y(t) + \ddot{y}(t) = 0$ with initial conditions $\dot{y}(0) = 1$ and $y(0) = 1$, for $t \geq 0$ is

A. 1 **B.** 2 **C.** π **D.** $\sqrt{2}$

Q6.11.1 MCQ 2011 2M

Consider the differential equation $\ddot{y} + 2\dot{y} + y = 0$ with boundary conditions $y(0) = 1$, $y(1) = 0$. The value of $y(2)$ is

A. -1 **B.** $-e^{-1}$ **C.** $-e^{-2}$ **D.** $-e^2$

Q6.10.1 MCQ 2010 2M

Consider the differential equation $\frac{dy}{dx} + y = e^x$ with $y(0) = 1$. The value of $y(1)$ is

A. $e + e^{-1}$ **B.** $\frac{1}{2}(e - e^{-1})$

C. $\frac{1}{2}(e + e^{-1})$ **D.** $2(e - e^{-1})$

Q6.08.1 MCQ 2008 2M

Consider the differential equation $\frac{dy}{dx} = 1 + y^2$. Which of the following can be a particular solution of this differential equation?

A. $y = \tan(x + 3)$ **B.** $y = \tan x + 3$

C. $x = \tan(y + 3)$ **D.** $x = \tan y + 3$

Q6.07.1 MCQ 2007 2M

The boundary-value problem $y'' + \lambda y = 0,\ y(0) = y(\pi) = 0$ will have non-zero solutions if and only if the values of λ are

A. $0, \pm 1, \pm 2, \ldots$ **B.** $1, 2, 3, \ldots$

C. $1, 4, 9, \ldots$ **D.** $1, 9, 25, \ldots$

Q6.06.1 MCQ 2006 2M

For an initial value problem $\ddot{y} + 2\dot{y} + 101y = 10.4e^x$, $y(0) = 1.1$ and $\dot{y} = -0.9$. Various solutions are written in the following groups. Match the type of solution with the correct expression.

	Group 1		Group 2
P.	General solution of homogeneous equations	1.	$0.1e^x$
		2.	$e^{-x}(A\cos 10x + B\sin 10x)$
Q.	Particular integral	3.	$e^{-x}\cos 10x + 0.1e^x$
R.	Total solution satisfying boundary conditions		

A. P–2, Q–1, R–3 **B.** P–1, Q–3, R–2

C. P–1, Q–2, R–3 **D.** P–3, Q–2, R–1

Q6.05.1 MCQ 2005 2M

The general solution of the differential equation $(D^2 - 4D + 4)y = 0$, is of the form (given $D = d/dx$), and C_1 and C_2 are constants

A. C_1e^{2x} **B.** $C_1e^{2x} + C_2e^{-2x}$

C. $C_1e^{2x} + C_2xe^{2x}$ **D.** $C_1e^{2x} + C_2xe^{-2x}$

Mechanical Engineering

Q6.26.1 MCQ 2026 1M

The order and degree of the following differential equation are m and n, respectively

$$\frac{\partial^3\varphi}{\partial x^3} + \frac{\partial^2\varphi}{\partial y^2}\frac{\partial\varphi}{\partial x} + \left(\frac{\partial^2\varphi}{\partial x^2}\right)^2 + \frac{\partial\varphi}{\partial y} = 0$$

The value of $(m - n)$ is

A. 2 **B.** 3 **C.** 1 **D.** 0

Q6.26.2 MCQ 2026 2M

Consider the following differential equation

$$\frac{\partial y}{\partial x} = 3\frac{\partial y}{\partial t} + y$$

If $y(x, 0) = 10e^{-2x}$, then the solution of the differential equation is

A. $y(x, t) = 10e^{-2x-t}$ **B.** $y(x, t) = 10e^{-2x+t}$

C. $y(x, t) = 10e^{-2x-2t}$ **D.** $y(x, t) = 10e^{-2x+2t}$

Q6.25.1 MCQ 2025 1M

For the differential equation given below, which one of the following options is correct?

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$0 \le x \le 1,\ 0 \le y \le 1$

A. $u = e^{x+y}$ is a solution for all x and y

B. $u = e^x \sin y$ is a solution for all x and y

C. $u = \sin x \sin y$ is a solution for all x and y

D. $u = \cos x \cos y$ is a solution for all x and y

Q6.25.2 NAT 2025 2M

Let y be the solution of the differential equation with the initial conditions given below. If $y(x = 2) = A\ln 2$, then the value of A is _____ (rounded off to 2 decimal places).

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0$$

$$y(x = 1) = 0 \qquad \frac{dy}{dx}(x = 1) = 1$$

Q6.24.1 NAT 2024 2M

If $x(t)$ satisfies the differential equation

$$t\frac{dx}{dt} + (t - x) = 0$$

subject to the condition $x(1) = 0$, then the value of $x(2)$ is _____ (rounded off to 2 decimal places)

Q6.23.1 NAT 2023 2M

Consider the second order linear ordinary differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$, $x \geq 1$ with initial condition $y(x = 1) = 6$, $\left.\frac{dy}{dx}\right|_{x=1} = 2$. The value of y at $x = 2$ equals _____ (Answer in integer).

Q6.22.1 MCQ 2022 1M

Solution of $\nabla^2 T = 0$ in a square domain $(0 < x < 1$ and $0 < y < 1)$ with boundary conditions:

$$T(x, 0) = x;\ T(0, y) = y;$$

$$T(x, 1) = 1 + x;\ T(1, y) = 1 + y$$

is

A. $T(x, y) = x - xy + y$

B. $T(x, y) = x + y$

C. $T(x, y) = -x + y$

D. $T(x, y) = x + xy + y$

Q6.22.2 MCQ 2022 2M

For the exact differential equation,

$$\frac{du}{dx} = \frac{-xu^2}{2 + x^2u}$$

Which one of the following is the solution?

A. $u^2 + 2x^2 = \text{Constant}$

B. $xu^2 + u = \text{Constant}$

C. $\frac{1}{2}x^2u^2 + 2u = \text{Constant}$

D. $\frac{1}{2}ux^2 + 2x = \text{Constant}$

Q6.21.1 MCQ 2021 1M

If $y(x)$ satisfies the differential equation

$$(\sin x)\frac{dy}{dx} + y\cos x = 1,$$

subject to the condition $y(\pi/2) = \pi/2$, then $y(\pi/6)$ is

A. 0 **B.** $\frac{\pi}{6}$ **C.** $\frac{\pi}{3}$ **D.** $\frac{\pi}{2}$

Q6.21.2 MCQ 2021 2M

Consider the following differential equation

$$(1 + y)\frac{dy}{dx} = y.$$

The solution of the equation that satisfies the condition $y(1) = 1$ is

A. $2ye^y = e^x + e$ **B.** $y^2e^y = e^x$

C. $ye^y = e^x$ **D.** $(1 + y)e^y = 2e^x$

Q6.19.1 MCQ 2019 1M

For the equation $\frac{dy}{dx} + 7x^2y = 0$, if $y(0) = 3/7$, then the value of $y(1)$ is

A. $\frac{7}{3}e^{-7/3}$ **B.** $\frac{7}{3}e^{-3/7}$ **C.** $\frac{3}{7}e^{-7/3}$ **D.** $\frac{3}{7}e^{-3/7}$

Q6.19.2 MCQ 2019 1M

The differential equation $\frac{dy}{dx} + 4y = 5$ is valid in the domain $0 \leq x \leq 1$ with $y(0) = 2.25$. The solution of the differential equation is

A. $y = e^{-4x} + 5$ **B.** $y = e^{-4x} + 1.25$

C. $y = e^{4x} + 5$ **D.** $y = e^{4x} + 1.25$

Q6.19.3 MCQ 2019 2M

A differential equation is given as

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 4.$$

The solution of the differential equation in terms of arbitrary constants C_1 and C_2 is

A. $y = C_1x^2 + C_2x + 2$ **B.** $y = C_1x^2 + C_2x + 4$

C. $y = \frac{C_1}{x^2} + C_2x + 2$ **D.** $y = \frac{C_1}{x^2} + C_2x + 4$

Q6.18.1 MCQ 2018 1M

If y is the solution of the differential equation $y^3\frac{dy}{dx} + x^3 = 0,\ y(0) = 1$, the value of $y(-1)$ is

A. -2 **B.** -1 **C.** 0 **D.** 1

Q6.18.2 NAT 2018 2M

Given the ordinary differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

with $y(0) = 0$ and $\frac{dy}{dx}(0) = 1$, the value of $y(1)$ is _____ (corrected to two decimal places).

Q6.17.1 MCQ 2017 1M

Consider the partial differential equation for $u(x, y)$ with the constant $c > 1$:

$$\frac{\partial u}{\partial y} + c\frac{\partial u}{\partial x} = 0$$

Solution of this equation is

A. $u(x, y) = f(x + cy)$ **B.** $u(x, y) = f(x - cy)$

C. $u(x, y) = f(cx + y)$ **D.** $u(x, y) = f(cx - y)$

Q6.17.2 MCQ 2017 1M

The differential equation $\frac{d^2y}{dx^2} + 16y = 0$ for $y(x)$ with the two boundary conditions $\left.\frac{dy}{dx}\right|_{x=0} = 1$ and $\left.\frac{dy}{dx}\right|_{x=\pi/2} = -1$ has

A. no solution **B.** exactly two solutions

C. exactly one solution **D.** infinitely many solutions

Q6.17.3 NAT 2017 2M

Consider the differential equation $3y''(x) + 27y(x) = 0$ with initial conditions $y(0) = 0$ and $y'(0) = 2000$. The value of y at $x = 1$ is _____.

Q6.16.1 NAT 2016 2M

If $y = f(x)$ satisfies the boundary value problem $y'' + 9y = 0.\ y(0) = 0,\ y(\pi/2) = \sqrt{2}$, then $y(\pi/4)$ is _____.

Q6.15.1 MCQ 2015 2M

Find the solution of $\frac{d^2y}{dx^2} = y$ which passes through the origin and the point $\left(\ln 2, \frac{3}{4}\right)$.

A. $y = (1/2)e^x - e^{-x}$ **B.** $y = (1/2)(e^x + e^{-x})$

C. $y = (1/2)(e^x - e^{-x})$ **D.** $y = (1/2)e^x + e^{-x}$

Q6.15.2 MCQ 2015 2M

Consider the following differential equation:

$$\frac{dy}{dt} = -5y;$$

initial condition: $y = 2$ at $t = 0$. The value of y at $t = 3$ is

A. $-5e^{-10}$ **B.** $2e^{-10}$ **C.** $2e^{-15}$ **D.** $-15e^2$

Q6.14.1 MCQ 2014 1M

The matrix form of the linear system $\frac{dx}{dt} = 3x - 5y$ and $\frac{dy}{dt} = 4x + 8y$

A. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

B. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 3 & 8 \\ 4 & -5 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

C. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 4 & -5 \\ 3 & 8 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

D. $\frac{d}{dt}\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{bmatrix} 4 & 8 \\ 3 & -5 \end{bmatrix}\begin{Bmatrix} x \\ y \end{Bmatrix}$

Q6.14.2 NAT 2014 2M

If $y = f(x)$ is the solution of $\frac{d^2y}{dx^2} = 0$ with the boundary conditions $y = 5$ at $x = 0$, and $\frac{dy}{dx} = 2$ at $x = 10$, $f(15) =$ _____

Q6.14.3 MCQ 2014 2M

The general solution of the differential equation $\frac{dy}{dx} = \cos(x + y)$, with c as a constant, is

A. $y + \sin(x + y) = x + c$

B. $\tan\left(\frac{x + y}{2}\right) = y + c$

C. $\cos\left(\frac{x + y}{2}\right) = x + c$

D. $\tan\left(\frac{x+y}{2}\right) = x + c$

Q6.14.4 MCQ 2014 2M

Consider two solutions $x(t) = x_1(t)$ and $x(t) = x_2(t)$ of the differential equation

$$\frac{d^2x(t)}{dt^2} + x(t) = 0,\ t > 0,$$

such that $x_1(0) = 1$, $\left.\frac{dx_1(t)}{dt}\right|_{t=0} = 0$, $x_2(0) = 0$, $\left.\frac{dx_2(t)}{dt}\right|_{t=0} = 1$. The Wronskian $W(t) = \begin{vmatrix} x_1(t) & x_2(t) \\ \frac{dx_1(t)}{dt} & \frac{dx_2(t)}{dt} \end{vmatrix}$ at $t = \pi/2$ is

A. 1 **B.** -1 **C.** 0 **D.** $\pi/2$

Q6.14.5 MCQ 2014 1M

The solution of the initial value problem $\frac{dy}{dx} = -2xy$; $y(0) = 2$ is

A. $1 + e^{-x^2}$ **B.** $2e^{-x^2}$ **C.** $1 + e^{x^2}$ **D.** $2e^{x^2}$

Q6.13.1 MCQ 2013 1M

The partial differential equation $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2}$ is a

A. Linear equation of order 2

B. Non-linear equation of order 1

C. Linear equation of order 1

D. Non-linear equation of order 2

Q6.13.2 MCQ 2013 2M

The solution to the differential equation $\frac{d^2u}{dx^2} - k\frac{du}{dx} = 0$ where k is a constant, subjected to the boundary conditions $u(0) = 0$ and $u(L) = U$, is

A. $u = U\frac{x}{L}$ **B.** $u = U\left(\frac{1 - e^{kx}}{1 - e^{kL}}\right)$

C. $u = U\left(\frac{1 - e^{-kx}}{1 - e^{-kL}}\right)$ **D.** $u = U\left(\frac{1 + e^{-kx}}{1 + e^{-kL}}\right)$

Q6.12.1 MCQ 2012 2M

Consider the differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 4y = 0$ with the boundary conditions of $y(0) = 0$ and $y(1) = 1$. The complete solution of the differential equation is

A. x^2 **B.** $\sin\left(\frac{\pi x}{2}\right)$

C. $e^x \sin\left(\frac{\pi x}{2}\right)$ **D.** $e^{-x} \sin\left(\frac{\pi x}{2}\right)$

Q6.11.1 MCQ 2011 2M

Consider the differential equation $\frac{dy}{dx} = (1 + y^2)x$. The general solution with constant c is

A. $y = \tan\frac{x^2}{2} + \tan c$ **B.** $y = \tan^2\left(\frac{x}{2} + c\right)$

C. $y = \tan^2\left(\frac{x^2}{2}\right) + c$ **D.** $y = \tan\left(\frac{x^2}{2} + c\right)$

Q6.10.1 MCQ 2010 1M

The Blasius equation $\frac{d^3f}{d\eta^3} + \frac{f}{2}\frac{d^2f}{d\eta^2} = 0$, is a

A. second order nonlinear ordinary differential equation

B. third order nonlinear ordinary differential equation

C. third order linear ordinary differential equation

D. mixed order nonlinear ordinary differential equation

Q6.09.1 MCQ 2009 2M

The solution of $x\frac{dy}{dx} + y = x^4$ with the condition $y(1) = \frac{6}{5}$ is

A. $y = \frac{x^4}{5} + \frac{1}{x}$ **B.** $y = \frac{4x^4}{5} + \frac{4}{5x}$

C. $y = \frac{x^4}{5} + 1$ **D.** $y = \frac{x^5}{5} + 1$

Q6.08.1 MCQ 2008 1M

Given that $\ddot{x} + 3x = 0$, and $x(0) = 1$, $\dot{x} = 0$, what is $x(1)$?

A. -0.99 **B.** -0.16 **C.** 0.16 **D.** 0.99

Q6.08.2 MCQ 2008 2M

Let $f = y^x$. What is $\frac{\partial^2 f}{\partial x \partial y}$ at $x = 2$, $y = 1$?

A. 0 **B.** $\ln 2$ **C.** 1 **D.** $\frac{1}{\ln 2}$

Q6.08.3 MCQ 2008 2M

It is given that $y'' + 2y' + y = 0,\ y(0) = 0,\ y(1) = 0$. What is $y(0.5)$?

A. 0 **B.** 0.37 **C.** 0.62 **D.** 1.13

Q6.07.1 MCQ 2007 1M

The partial differential equation

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial y} = 0$$

has

A. degree 1 order 2 **B.** degree 1 order 1

C. degree 2 order 1 **D.** degree 2 order 2

Q6.07.2 MCQ 2007 2M

The solution of $\frac{dy}{dx} = y^2$ with initial value $y(0) = 1$ bounded in the interval

A. $-\infty \le x \le \infty$ **B.** $-\infty \le x \le 1$

C. $x < 1,\ x > 1$ **D.** $-2 \le x \le 2$

Q6.06.1 MCQ 2006 1M

The solution of the differential equation $\frac{dy}{dx} + 2xy = e^{-x^2}$ with $y(0) = 1$ is

A. $(1 + x)e^{+x^2}$ **B.** $(1 + x)e^{-x^2}$

C. $(1 - x)e^{+x^2}$ **D.** $(1 - x)e^{-x^2}$

Q6.06.2 MCQ 2006 2M

For $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 3e^{2x}$, the particular integral is:

A. $\frac{1}{15}e^{2x}$ **B.** $\frac{1}{5}e^{2x}$

C. $3e^{2x}$ **D.** $C_1e^{-x} + C_2e^{-3x}$

Q6.05.1 MCQ 2005 2M

If $x^2\frac{dy}{dx} + 2xy = \frac{2\ln x}{x}$, and $y(1) = 0$, then what is $y(e)$?

A. e **B.** 1 **C.** $\frac{1}{e}$ **D.** $\frac{1}{e^2}$

Common Data Questions

Answer the next two questions using the following data. The complete solution for the ordinary differential equation $\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = 0$ is $y = c_1e^{-x} + c_2e^{-3x}$

Q6.05.2 MCQ 2005 2M

Then, p and q are

A. $p = 3,\ q = 3$ **B.** $p = 3,\ q = 4$

C. $p = 4,\ q = 3$ **D.** $p = 4,\ q = 4$

Q6.05.3 MCQ 2005 2M

Which of the following is a solution of the differential equation

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + (q + 1)y = 0?$$

A. e^{-3x} **B.** xe^{-x} **C.** xe^{-2x} **D.** x^2e^{-2x}

Q6.03.1 MCQ 2003 2M

The solution of the differential equation $\frac{dy}{dx} + y^2 = 0$ is

A. $y = \frac{1}{x + c}$

B. $y = \frac{-x^3}{3} + c$

C. ce^x

D. unsolvable as equation is non-linear

Q6.01.1 MCQ 2001 5M

Solve the differential equation

$$\frac{d^2y}{dx^2} + y = x$$

with the following conditions:

(i) at $x = 0,\ y = 1$

(ii) at $x = \frac{\pi}{2},\ y = \frac{\pi}{2}$

A. $\cos x + x^2$ **B.** $\cos x - x$

C. $\cos x + x$ **D.** $\sin x + x^2$

Q6.00.1 MCQ 2000 1M

The solution of the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$ is

A. $Ae^x + Be^{-x}$

B. $e^x(Ax + B)$

C. $e^{-x}\left\{A\cos\left(\frac{\sqrt{3}}{2}\right)x + B\cos\left(\frac{\sqrt{3}}{2}\right)x\right\}$

D. $e^{-x/2}\left\{A\cos\left(\frac{\sqrt{3}}{2}\right)x + B\sin\left(\frac{\sqrt{3}}{2}\right)x\right\}$

Q6.99.1 MCQ 1999 2M

$$\frac{d^2y}{dx^2} + (x^2 + 4x)\frac{dy}{dx} + y = x^8 - 8$$

The above equation is a

A. partial differential equation

B. nonlinear differential equation

C. non-homogeneous differential equation

D. ordinary differential equation

Q6.98.1 MCQ 1998 2M

The general solution of the differential equation $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ is:

A. $Ax + Bx^2$ (A, B are constants)

B. $Ax + B\log x$ (A, B are constants)

C. $Ax + Bx^2\log x$ (A, B are constants)

D. $Ax + Bx\log x$ (A, B are constants)

Q6.98.2 NAT 1998 5M

The radial displacement in a rotating disc is governed by the differential equation

$$\frac{d^2u}{dx^2} + \frac{1}{x}\frac{du}{dx} - \frac{u}{x^2} = 8x$$

where u is the displacement and x is the radius.

If $u = 0$ at $x = 0$, and $u = 2$ at $x = 1$, calculate the displacement at $x = \frac{1}{2}$

Q6.96.1 MCQ 1996 2M

The particular solution for the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 5\cos x \text{ is:}$$

A. $0.5\cos x + 1.5\sin x$ **B.** $1.5\cos x + 0.5\sin x$

C. $1.5\sin x$ **D.** $0.5\cos x$

Q6.95.1 MCQ 1995 1M

A differential equation of form $\frac{dy}{dx} = f(x, y)$ is homogeneous if the function $f(x, y)$ depends only on the ratio $\frac{y}{x}$ or $\frac{x}{y}$

A. True

B. False

Q6.95.2 MCQ 1995 1M

The solution to the differential equation $f''(x) + 4f'(x) + 4f(x) = 0$ is:

A. $f_1(x) = e^{-2x}$

B. $f_1(x) = e^{2x}$, $f_2(x) = e^{-2x}$

C. $f_1(x) = e^{-2x}$, $f_2(x) = xe^{-2x}$

D. $f_1(x) = e^{-2x}$, $f_2(x) = e^{-x}$

Q6.94.1 MCQ 1994 1M

For the differential equation $\frac{dy}{dx} + 5y = 0$ with $y(0) = 1$, the general solution is

A. e^{5t} **B.** e^{-5t} **C.** $5e^{-5t}$ **D.** $e^{\sqrt{-5t}}$

Q6.94.2 MCQ 1994 1M

If $H(x, y)$ is a homogeneous function of degree n, then $x\frac{\partial H}{\partial x} + y\frac{\partial H}{\partial y} = nH$.

A. True

B. False

Q6.94.3 MCQ 1994 2M

Solve for y, if $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$ with $y(0) = 1$ and $y'(0) = 2$.

A. $2e^{-t} + te^{-t}$ **B.** $e^{-t} + 2te^{-2t}$

C. $e^{-t} + 3te^{-2t}$ **D.** $e^{-t} + 3te^{-t}$

Q6.93.1 MCQ 1993 1M

The differential equation $y'' + y = 0$ is subjected to the conditions $y(0) = 0$, $y(\lambda) = 0$. In order that the equation has non-trivial solutions, the general value of λ is

A. $\frac{m\pi}{2}$ **B.** $m\pi$ **C.** $\frac{m\pi}{4}$ **D.** $\frac{m\pi}{6}$

Q6.93.2 MCQ 1993 1M

The differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + \sin y = 0$ is

A. Linear **B.** Non-linear

C. Homogeneous **D.** of degree 2

Civil Engineering

Q6.26.1 MCQ 2026 2M

An ordinary differential equation is given below.

$$x^2\frac{d^2y}{dx^2} = 6y$$

Considering a and b as arbitrary constants, the general solution of the equation is

A. $y(x) = ax^3 + \frac{b}{x^2}$ **B.** $y(x) = ax^2 + \frac{b}{x^3}$

C. $y(x) = ax^2 + b\ln x$ **D.** $y(x) = ax^3 + b\ln x$

Q6.26.2 MSQ 2026 1M

A partial differential equation is given below.

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$$

Possible solutions(s) is/are:

☐ **A.** $(x + y)^5$

☐ **B.** $(x - 2y)^3$

☐ **C.** $\cos(x + y)$

☐ **D.** $\sin(x - 2y)$

Q6.26.3 NAT 2026 2M

Consider differential equation $\frac{dy}{dx} + xy = x$ with the condition as $y = 0$ at $x = 0$. The value of y at $x = 1.0$ is _____ *(rounded off to two decimal places).*

Q6.25.1 MSQ 2025 1M

Which of the following equations belong/belongs to the class of second-order, linear, homogeneous partial differential equations:

☐ **A.** $\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + xy$

☐ **B.** $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

☐ **C.** $\frac{\partial u}{\partial t} = c\frac{\partial u}{\partial x}$

☐ **D.** $\left(\frac{\partial^2 u}{\partial t^2}\right)^2 = c^2\frac{\partial^2 u}{\partial x^2}$

Q6.25.2 NAT 2025 2M

Let y be the solution of the initial value problem $y'' + 0.8y' + 0.16y = 0$, where $y(0) = 3$ and $y'(0) = 4.5$. Then, $y(1)$ is equal to _____ *(rounded off to 1 decimal place).*

Q6.25.3 NAT 2025 1M

The "order" of the following ordinary differential equation is _____.

$$\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^6 + \left(\frac{dy}{dx}\right)^4 + y = 0$$

Q6.25.4 MCQ 2025 2M

Pick the **CORRECT** solution for the following differential equation

$$\frac{dy}{dx} = e^{x-y}$$

A. $y = \ln(e^x + \text{Constant})$

B. $\ln(y) = x + \text{Constant}$

C. $\ln(y) = \ln(e^x) + \text{Constant}$

D. $y = x + \text{Constant}$

Q6.24.1 MCQ 2024 1M

The second-order differential equation in an unknown function $u : u(x, y)$ is

$$\frac{\partial^2 u}{\partial x^2} = 2$$

Assuming $g : g(x)$, $f : f(y)$, and $h : h(y)$, the general solution of the above differential equation is

A. $u = x^2 + f(y) + g(x)$

B. $u = x^2 + xf(y) + h(y)$

C. $u = x^2 + xf(y) + g(x)$

D. $u = x^2 + f(y) + y\,g(x)$

Q6.24.2 MSQ 2024 1M

For the following partial differential equation,

$$x\frac{\partial^2 f}{\partial x^2} + y\frac{\partial^2 f}{\partial y^2} = \frac{x^2 + y^2}{2}$$

which of the following option(s) is/are CORRECT?

- **A.** elliptic for $x > 0$ and $y > 0$
- **B.** parabolic for $x > 0$ and $y > 0$
- **C.** elliptic for $x = 0$ and $y > 0$
- **D.** hyperbolic for $x < 0$ and $y > 0$

Q6.24.3 MCQ 2024 1M

Consider two Ordinary Differential Equations (ODEs):

P: $\dfrac{dy}{dx} = \dfrac{x^4 + 3x^2y^2 + 2y^4}{x^3y}$

Q: $\dfrac{dy}{dx} = \dfrac{-y^2}{x^2}$

Which one of the following options is CORRECT?

- **A.** P is a homogeneous ODE and Q is an exact ODE
- **B.** P is a homogeneous ODE and Q is not an exact ODE
- **C.** P is a nonhomogeneous ODE and Q is an exact ODE
- **D.** P is a nonhomogeneous ODE and Q is not an exact ODE

Q6.23.1 NAT 2023 1M

In the differential equation $\dfrac{dy}{dx} + \alpha xy = 0$, α is a positive constant. If $y = 1.0$ at $x = 0.0$, and $y = 0.8$ at $x = 1.0$, the value of α is _____ (rounded off to three decimal places).

Q6.23.2 MCQ 2023 2M

The solution of the differential equation

$$\frac{d^3y}{dx^3} - 5.5\frac{d^2y}{dx^2} + 9.5\frac{dy}{dx} - 5y = 0$$

is expressed as $y = C_1e^{2.5x} + C_2e^{\alpha x} + C_3e^{\beta x}$, where C_1, C_2, C_3, α, and β are constants, with α and β being distinct and not equal to 2.5. Which of the following options is correct for the values of α and β ?

A. 1 and 2 **B.** -1 and -2

C. 2 and 3 **D.** -2 and -3

Q6.22.1 MCQ 2022 1M

Consider the following expression:

$$z = \sin(y + it) + \cos(y - it)$$

where z, y, and t are variables, and $i = \sqrt{-1}$ is a complex number. The partial differential equation derived from the above expression is

A. $\dfrac{\partial^2 z}{\partial t^2} + \dfrac{\partial^2 z}{\partial y^2} = 0$ **B.** $\dfrac{\partial^2 z}{\partial t^2} - \dfrac{\partial^2 z}{\partial y^2} = 0$

C. $\dfrac{\partial z}{\partial t} - i\dfrac{\partial z}{\partial y} = 0$ **D.** $\dfrac{\partial z}{\partial t} + i\dfrac{\partial z}{\partial y} = 0$

Q6.22.2 MCQ 2022 1M

For the equation

$$\frac{d^3y}{dx^3} + x\left(\frac{dy}{dx}\right)^{3/2} + x^2y = 0$$

the correct description is

- **A.** an ordinary differential equation of order 3 and degree 2
- **B.** an ordinary differential equation of order 3 and degree 3
- **C.** an ordinary differential equation of order 2 and degree 3
- **D.** an ordinary differential equation of order 3 and degree 3/2

Q6.21.1 MCQ 2021 2M

The solution of the second-order differential equation $\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} + y = 0$ with boundary conditions $y(0) = 1$ and $y(1) = 3$ is

- **A.** $e^{-x} + (3e - 1)xe^{-x}$
- **B.** $e^{-x} - (3e - 1)xe^{-x}$
- **C.** $e^{-x} + \left[3e\sin\left(\dfrac{\pi x}{2}\right) - 1\right]xe^{-x}$
- **D.** $e^{-x} - \left[3e\sin\left(\dfrac{\pi x}{2}\right) - 1\right]xe^{-x}$

Q6.21.2 MCQ 2021 2M

If k is a constant, the general solution of $\frac{dy}{dx} - \frac{y}{x} = 1$ will be in the form of

A. $y = x\ln(kx)$ **B.** $y = k\ln(kx)$

C. $y = x\ln(x)$ **D.** $y = xk\ln(k)$

Q6.20.1 MCQ 2020 1M

In the following partial differential equation, θ is a function of 't' and 'z' and 'D' and 'K' are functions of θ

$$D(\theta)\frac{\partial^2\theta}{\partial z^2} + \frac{\partial K(\theta)}{\partial z} - \frac{\partial\theta}{\partial t} = 0$$

The above equation is

A. a second order linear equation

B. a second order non-linear equation

C. a second-degree non-linear equation

D. a second-degree linear equation

Q6.20.2 MCQ 2020 2M

For the ordinary differential equation $\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$, with initial condition $x(0) = 0$ and $\frac{dx}{dt}(0) = 10$, the solution is

A. $-10\,e^{2t} + 10\,e^{3t}$ **B.** $5\,e^{2t} + 6\,e^{3t}$

C. $10\,e^{2t} + 10\,e^{3t}$ **D.** $-5\,e^{2t} + 6\,e^{3t}$

Q6.20.3 MCQ 2020 1M

The ordinary differential equation $\frac{d^2u}{dx^2} - 2x^2u + \sin x = 0$ is

A. linear and homogeneous

B. non-linear and homogeneous

C. non-linear and nonhomogeneous

D. linear and nonhomogeneous

Q6.20.4 MCQ 2020 1M

The following partial differential equation is defined for $u : u(x, y)$

$$\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial x^2};\ y \geq 0;\ x_1 \leq x \leq x_2$$

The set of auxiliary conditions necessary to solve the equation uniquely, is

A. one initial condition and two boundary conditions

B. three initial conditions

C. two initial conditions and one boundary condition

D. three boundary conditions

Q6.20.5 MCQ 2020 2M

An ordinary differential equation is given below:

$$6\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$$

The general solution of the above equation (with constants C_1 and C_2), is

A. $y(x) = C_1\,e^{x/3} + C_2\,e^{-x/2}$

B. $y(x) = C_1x\,e^{-x/3} + C_2\,e^{x/2}$

C. $y(x) = C_1\,e^{-x/3} + C_2\,e^{x/2}$

D. $y(x) = C_1\,e^{-x/3} + C_2x\,e^{x/2}$

Q6.19.1 MCQ 2019 1M

Consider a two-dimensional flow through isotropic soil along x direction and z direction. If h is the hydraulic head, the Laplace's equation of continuity is expressed as

A. $\frac{\partial h}{\partial x} + \frac{\partial h}{\partial z} = 0$ **B.** $\frac{\partial h}{\partial x} + \frac{\partial h}{\partial x}\frac{\partial h}{\partial z} + \frac{\partial h}{\partial z} = 0$

C. $\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial z^2} = 0$ **D.** $\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial x\,\partial z} + \frac{\partial^2 h}{\partial z^2} = 0$

Q6.19.2 NAT 2019 2M

Consider the ordinary differential equation $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$. Given the values of $y(1) = 0$ and $y(2) = 2$, the value of $y(3)$ (round off to 1 decimal place), is _____

Q6.19.3 MCQ 2019 2M

An ordinary differential equation is given below.

$$\left(\frac{dy}{dx}\right)(x\ln x) = y$$

The solution for the above equation is (*Note: K denotes a constant in the options*)

A. $y = Kx\ln x$ **B.** $y = Kxe^x$

C. $y = Kxe^{-x}$ **D.** $y = K\ln x$

Q6.18.1 MCQ 2018 2M

The solution at $x = 1, t = 1$ of the partial differential equation $\frac{\partial^2 u}{\partial x^2} = 25\frac{\partial^2 u}{\partial t^2}$ subject to initial conditions of $u(0) = 3x$ and $\frac{\partial u}{\partial t}(0) = 3$ is _____

A. 1 **B.** 2 **C.** 4 **D.** 6

Q6.18.2 NAT 2018 2M

The solution (up to three decimal places) at $x = 1$ of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ subject to boundary conditions $y(0) = 1$ and $\frac{dy}{dx}(0) = -1$ is _____

Q6.18.3 MCQ 2018 1M

The solution of the equation $x\frac{dy}{dx} + y = 0$ passing through the point $(1, 1)$ is

A. x **B.** x^2 **C.** x^{-1} **D.** x^{-2}

Q6.17.1 NAT 2017 1M

Consider the following partial differential equation

$$3\frac{\partial^2\phi}{\partial x^2} + B\frac{\partial^2\phi}{\partial x\partial y} + 3\frac{\partial^2\phi}{\partial y^2} + 4\phi = 0$$

For this equation to be classified as parabolic, the value of B^2 must be _____

Q6.17.2 MCQ 2017 2M

The solution of the equation $\frac{dQ}{dt} + Q = 1$ with $Q = 0$ at $t = 0$ is

A. $Q(t) = e^{-t} - 1$ **B.** $Q(t) = 1 + e^{-t}$

C. $Q(t) = 1 - e^{t}$ **D.** $Q(t) = 1 - e^{-t}$

Q6.17.3 MCQ 2017 2M

Consider the following second-order differential equation:

$$y'' - 4y' + 3y = 2t - 3t^2$$

The particular solution of the differential equation is

A. $-2 - 2t - t^2$ **B.** $-2t - t^2$

C. $2t - 3t^2$ **D.** $-2 - 2t - 3t^2$

Q6.16.1 MCQ 2016 1M

The type of partial differential equation $\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + 3\frac{\partial^2 P}{\partial x\partial y} + 2\frac{\partial P}{\partial x} - \frac{\partial P}{\partial y} = 0$ is

A. elliptic **B.** parabolic

C. hyperbolic **D.** none of these

Q6.16.2 MCQ 2016 1M

The solution of the partial differential equation $\frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2}$ is of the form

A. $C\cos(kt)\left[C_1 e^{(\sqrt{k/\alpha})x} + C_2 e^{-(\sqrt{k/\alpha})x}\right]$

B. $C e^{kt}\left[C_1 e^{(\sqrt{k/\alpha})x} + C_2 e^{-(\sqrt{k/\alpha})x}\right]$

C. $C e^{kt}\left[C_1 \cos(\sqrt{k/\alpha})x + C_2 \sin(-\sqrt{k/\alpha})x\right]$

D. $C\sin(kt)\left[C_1 \cos(\sqrt{k/\alpha})x + C_2 \sin(-\sqrt{k/\alpha})x\right]$

Q6.16.3 MCQ 2016 2M

The respective expressions for complimentary function and particular integral part of the solution of the differential equation $\frac{d^4y}{dx^4} + 3\frac{d^2y}{dx^2} = 108x^2$ are

A. $[c_1 + c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[3x^4 - 12x^2 + c]$

B. $[c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[5x^4 - 12x^2 + c]$

C. $[c_1 + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[3x^4 - 12x^2 + c]$

D. $[c_1 + c_2x + c_3\sin\sqrt{3}x + c_4\cos\sqrt{3}x]$ and $[5x^4 - 12x^2 + c]$

Q6.15.1 MCQ 2015 2M

Consider the following differential equation:

$$x(y\,dx + x\,dy)\cos\frac{y}{x} = y(x\,dy - y\,dx)\sin\frac{y}{x}$$

Which of the following is the solution of the above equation (c is an arbitrary constant)?

A. $\frac{x}{y}\cos\frac{y}{x} = c$ **B.** $\frac{x}{y}\sin\frac{y}{x} = c$

C. $xy\cos\frac{y}{x} = c$ **D.** $xy\sin\frac{y}{x} = c$

Q6.15.2 NAT 2015 2M

Consider the following second order linear differential equation

$$\frac{d^2y}{dx^2} = -12x^2 + 24x - 20$$

The boundary conditions are: at $x = 0$, $y = 5$ and at $x = 2$, $y = 21$. The value of y at $x = 1$ is _____.

Q6.14.1 MCQ 2014 1M

The integrating factor for the differential equation $\frac{dP}{dt} + k_2P = k_1L_0e^{-k_1t}$ is

A. e^{-k_1t} **B.** e^{-k_2t} **C.** e^{k_1t} **D.** e^{k_2t}

Q6.14.2 NAT 2014 2M

Water is flowing at a steady rate through a homogeneous and saturated horizontal soil strip of 10m length. The strip is being subjected to a constant water head (H) of 5m at the beginning and 1m at the end. If the governing equation of flow in the soil strip is $\frac{d^2H}{dx^2} = 0$ (where x is the distance along the soil strip), the value of H (in m) at the middle of the strip is _____.

Q6.13.1 NAT 2013 2M

Laplace equation for water flow in soil is given by

$$\frac{\partial^2H}{\partial x^2} + \frac{\partial^2H}{\partial y^2} + \frac{\partial^2H}{\partial z^2} = 0$$

Head H does not vary in y and z-direction. Boundary condition $x = 0$, $H = 5m$, $\frac{dH}{dx} = -1$. What is the value of H at $x = 1.2$?

Q6.12.1 MCQ 2012 2M

The solution of the ordinary differential equation $\frac{dy}{dx} + 2y = 0$ for the boundary condition, $y = 5$ at $x = 1$ is

A. $y = e^{-2x}$ **B.** $y = 2e^{-2x}$

C. $y = 10.95e^{-2x}$ **D.** $y = 36.95e^{-2x}$

Q6.11.1 MCQ 2011 2M

The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = x$, with the condition that $y = 1$ at $x = 1$, is

A. $y = \frac{2}{3x^2} + \frac{x}{3}$ **B.** $y = \frac{x}{2} + \frac{1}{2x}$

C. $y = \frac{2}{3} + \frac{x}{3}$ **D.** $y = \frac{2}{3x} + \frac{x^2}{3}$

Q6.10.1 MCQ 2010 1M

The order and degree of the differential equation

$$\frac{d^3y}{dx^3} + 4\sqrt{\left(\frac{dy}{dx}\right)^3 + y^2} = 0$$

are respectively

A. 3 and 2 **B.** 2 and 3

C. 3 and 3 **D.** 3 and 1

Q6.10.2 MCQ 2010 2M

The solution to the ordinary differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$ is

A. $y = c_1e^{3x} + c_2e^{-2x}$ **B.** $y = c_1e^{3x} + c_2e^{2x}$

C. $y = c_1e^{-3x} + c_2e^{2x}$ **D.** $y = c_1e^{-3x} + c_2e^{-2x}$

Q6.10.3 MCQ 2010 2M

The partial differential equation that can be formed from $z = ax + by + ab$ has the form $\left(\text{with } p = \frac{\partial z}{\partial x} \text{ and } q = \frac{\partial z}{\partial y}\right)$

A. $z = px + qy$ **B.** $z = px + pq$

C. $z = px + qy + pq$ **D.** $z = qy + pq$

Q6.09.1 MCQ 2009 2M

Solution of the differential equation $3y\frac{dy}{dx} + 2x = 0$ represent a family of

A. ellipses **B.** circles

C. parabolas **D.** hyperbolas

Q6.08.1 MCQ 2008 1M

The general solution of $\frac{d^2y}{dx^2} + y = 0$ is

A. $y = P\cos x + Q\sin x$ **B.** $y = P\cos x$

C. $y = P\sin x$ **D.** $y = P\sin^2 x$

Q6.08.2 MCQ 2008 2M

The equation $k_x\frac{\partial^2h}{\partial x^2} + k_z\frac{\partial^2h}{\partial z^2} = 0$ can be transformed to $\frac{\partial^2h}{\partial x_1^2} + \frac{\partial^2h}{\partial z^2} = 0$ by substituting

A. $x_1 = x\frac{k_z}{k_x}$ **B.** $x_1 = x\frac{k_x}{k_z}$

C. $x_1 = x\sqrt{\frac{k_x}{k_z}}$ **D.** $x_1 = x\sqrt{\frac{k_z}{k_x}}$

Q6.08.3 MCQ 2008 2M

Solution of $\frac{dy}{dx} = -\frac{x}{y}$ at $x = 1$ and $y = \sqrt{3}$ is

A. $x - y^2 = 2$ **B.** $x + y^2 = 4$

C. $x^2 - y^2 = -2$ **D.** $x^2 + y^2 = 4$

Q6.07.1 MCQ 2007 1M

The degree of the differential equation $\frac{d^2x}{dt^2} + 2x^3 = 0$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q6.07.2 MCQ 2007 1M

The solution for the differential equation $\frac{dy}{dx} = x^2y$ with condition that $y = 1$ at $x = 0$ is

A. $y = e^{1/2x}$ **B.** $\ln(y) = \frac{x^3}{3} + 4$

C. $\ln(y) = \frac{x^2}{2}$ **D.** $y = e^{x^3/3}$

Q6.06.1 MCQ 2006 2M

The solution of the differential equation, $x^2\frac{dy}{dx} + 2xy - x + 1 = 0$, given that at $x = 1$, $y = 0$ is

A. $\frac{1}{2} - \frac{1}{x} + \frac{1}{2x^2}$ **B.** $\frac{1}{2} - \frac{1}{x} - \frac{1}{2x^2}$

C. $\frac{1}{2} + \frac{1}{x} + \frac{1}{2x^2}$ **D.** $-\frac{1}{2} + \frac{1}{x} + \frac{1}{2x^2}$

Q6.05.1 MCQ 2005 2M

Transformation to linear form by substituting $v = y^{1-n}$ of the equation $\frac{dy}{dt} + p(t)y = q(t)\,y^n$, $n > 0$ will be

A. $\frac{dv}{dt} + (1-n)pv = (1-n)q$

B. $\frac{dv}{dt} + (1-n)pv = (1+n)q$

C. $\frac{dv}{dt} + (1+n)pv = (1-n)q$

D. $\frac{dv}{dt} + (1+n)pv = (1+n)q$

Q6.05.2 MCQ 2005 2M

The solution of $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 17y = 0$; $y(0) = 1$, $\frac{dy}{dx}\left(\frac{\pi}{4}\right) = 0$ in the range $0 < x < \pi/4$ is given by

A. $e^{-x}\left(\cos 4x + \frac{1}{4}\sin 4x\right)$

B. $e^{x}\left(\cos 4x - \frac{1}{4}\sin 4x\right)$

C. $e^{-4x}\left(\cos x - \frac{1}{4}\sin x\right)$

D. $e^{-4x}\left(\cos 4x - \frac{1}{4}\sin 4x\right)$

Q6.04.1 MCQ 2004 2M

Biotransformation of an organic compound having concentration (x) can be modeled using an ordinary differential equation $\frac{dx}{dt} + kx^2 = 0$, where k is the reaction rate constant. If $x = a$ at $t = 0$, the solution of the equation is

A. $x = ae^{-kt}$ **B.** $\frac{1}{x} = \frac{1}{a} + kt$

C. $x = a(1 - e^{-kt})$ **D.** $x = a + kt$

Q6.01.1 MCQ 2001 1M

The number of boundary conditions required to solve the differential equation $\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$ is

A. 2 **B.** 0 **C.** 4 **D.** 1

Q6.01.2 MCQ 2001 2M

The solution for the following differential equation with boundary conditions $y(0) = 2$ and $y'(1) = -3$ is

$$\frac{d^2y}{dx^2} = 3x - 2$$

A. $y = \frac{x^3}{3} - \frac{x^2}{2} + 3x - 6$

B. $y = 3x^3 - \frac{x^2}{2} - 5x + 2$

C. $y = \frac{x^3}{2} - x^2 - \frac{5x}{2} + 2$

D. $y = x^3 - \frac{x^2}{2} + 5x + \frac{3}{2}$

Q6.00.1 MCQ 2000 2M

If $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$, $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ is equal to

A. Zero **B.** 1

C. 2 **D.** $-3(x^2 + y^2 + z^2)^{5/2}$

Q6.99.1 MCQ 1999 2M

If c is a constant, solution of the equation $\frac{dy}{dx} = 1 + y^2$ is

A. $y = \sin(x + c)$ **B.** $y = \cos(x + c)$

C. $y = \tan(x + c)$ **D.** $y = e^x + c$

Q6.98.1 MCQ 1998 5M

Solve $\frac{d^4y}{dx^4} - y = 15\cos 2x$

A. $C_1e^x + C_2e^{-x} + C_3\cos x + C_4\sin x + \cos 2x$

B. $C_1e^{2x} + C_2e^{-x} + C_3\cos x + C_4\sin x + \cos x$

C. $C_1e^{2x} + C_2e^{-2x} + C_3\cos x + C_4\sin x + \cos 2x$

D. $C_1e^{-2x} + C_2e^{-x} + C_3\cos x + C_4\sin x + \cos x$

Q6.97.1 MCQ 1997 1M

For the differential equation, $f(x, y)\frac{dy}{dx} + g(x, y) = 0$ to be exact,

A. $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$ **B.** $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}$

C. $f = g$ **D.** $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 g}{\partial y^2}$

Q6.97.2 MCQ 1997 1M

The differential equation $\frac{dy}{dx} + Py = Q$, is a linear equation of first order only if

A. P is a constant but Q is a function of y

B. P and Q are functions of y or constants

C. P is a function of y but Q is a constant

D. P and Q are functions of x or constants

Q6.95.1 MCQ 1995 2M

The solution of a differential equation $y'' + 3y' + 2y = 0$ is of the form

A. $c_1e^x + c_2e^{2x}$ **B.** $c_1e^{-x} + c_2e^{3x}$

C. $c_1e^{-x} + c_2e^{-2x}$ **D.** $c_1e^{x-2} + c_2 2^{-x}$

Q6.95.2 MCQ 1995 1M

The differential equation $y'' + (x^3 \sin x)^5 y' + y = \cos x^3$ is

A. homogeneous

B. non-linear

C. second order linear

D. non-homogeneous with constant coefficients

Q6.94.1 MCQ 1994 2M

The differential equation $\frac{d^4y}{dx^4} + P\frac{d^2y}{dx^2} + ky = 0$ is

A. Linear of fourth order

B. Non-Linear of fourth order

C. Non-Homogeneous

D. Linear and fourth degree

Q6.94.2 MCQ 1994 1M

The necessary and sufficient condition for the differential equation of the form $M(x, y)\,dx + N(x, y)\,dy = 0$ to be exact is

A. $M = N$ **B.** $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

C. $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ **D.** $\frac{\partial^2 M}{\partial x^2} = \frac{\partial^2 N}{\partial y^2}$

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Q6.26.1 NAT 2026 2M

Consider a differential equation

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} - 3y = 0$$

$$\left.\begin{aligned} y &= 3 \\ \frac{dy}{dx} &= -5 \end{aligned}\right\} \text{ at } x = 1$$

The value of y at $x = 2$ is _____ (*rounded off to two decimal places*).

Q6.25.1 MSQ 2025 2M

Consider the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = 0$$

Choose the CORRECT option(s) for the solution of y.

- [ ] **A.** $y = x + c$; c is a constant
- [ ] **B.** $y = \frac{c}{x}$; c is a constant
- [ ] **C.** $y = -x + c$; c is a constant
- [ ] **D.** $y = 0$

Q6.24.1 NAT 2024 2M

Consider the ordinary differential equation

$$x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = 0,$$

with the boundary conditions $y(x = 1) = 2$ and $y(x = 2) = 17/2$. The solution $y(x)$ at $x = 3/2$, rounded off to 2 decimal places, is _____

Q6.23.1 NAT 2023 1M

The position $x(t)$ of a particle, at constant ω, is described by the equation

$$\frac{d^2x}{dt^2} = -\omega^2 x.$$

The initial conditions are $x(t = 0) = 1$ and $\left.\frac{dx}{dt}\right|_{t=0} = 0$.

The position of the particle at $t = (3\pi/\omega)$ is _____ (in integer).

Q6.22.1 MCQ 2022 2M

The partial differential equation

$$\frac{\partial u}{\partial t} = \frac{1}{\pi^2}\frac{\partial^2 u}{\partial x^2}$$

where, $t \geq 0$ and $x \in [0, 1]$, is subjected to the following initial and boundary conditions:

$$u(x, 0) = \sin(\pi x)$$

$$u(0, t) = 0$$

$$u(1, t) = 0$$

The value of t at which $\frac{u(0.5, t)}{u(0.5, 0)} = \frac{1}{e}$ is

A. 1 **B.** e **C.** π **D.** $1/e$

Q6.21.1 NAT 2021 2M

For the ordinary differential equation

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y = 1$$

with the initial conditions $y(0) = y'(0) = y''(0) = y'''(0) = 0$, the value of $\lim_{t\to\infty} y(t) =$ _____ (round off to 3 decimal places).

Q6.20.1 NAT 2020 2M

Given $\frac{dy}{dx} = y - 20$, and $y|_{x=0} = 40$, the value of y at $x = 2$ is _____ (round off to nearest integer).

Q6.19.1 MCQ 2019 2M

The solution of the ordinary differential equation $\frac{dy}{dx} + 3y = 1$, subject to the initial condition $y = 1$ at $x = 0$, is

A. $\frac{1}{3}(1 + 2e^{-x/3})$ **B.** $\frac{1}{3}(5 - 2e^{-x/3})$

C. $\frac{1}{3}(5 - 2e^{-3x})$ **D.** $\frac{1}{3}(1 + 2e^{-3x})$

Q6.18.1 MCQ 2018 1M

Consider the following two equations:

$$\frac{dx}{dt} + x + y = 0$$

$$\frac{dy}{dt} - x = 0$$

The above set of equations is represented by

A. $\frac{d^2y}{dt^2} - \frac{dy}{dt} - y = 0$ **B.** $\frac{d^2x}{dt^2} - \frac{dx}{dt} - y = 0$

C. $\frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0$ **D.** $\frac{d^2x}{dt^2} + \frac{dx}{dt} + y = 0$

Q6.17.1 NAT 2017 2M

For the initial value problem

$$\frac{dx}{dt} = \sin(t),\ x(0) = 0$$

the value of x at $t = \pi/3$, is _____.

Q6.16.1 MCQ 2016 2M

What is the solution for the second order differential equation $\frac{d^2y}{dx^2} + y = 0$, with the initial condition $y|_{x=0} = 5$ and $\frac{dy}{dx}\Big|_{x=0} = 10$?

A. $y = 5 + 10\sin x$ **B.** $y = 5\cos x - 5\sin x$

C. $y = 5\cos x + 10x$ **D.** $y = 5\cos x + 10\sin x$

Q6.15.1 MCQ 2015 1M

Consider a linear ordinary differential equation: $\frac{dy}{dx} + p(x)y = r(x)$. Functions $p(x)$ and $r(x)$ are defined and have a continuous first derivative. The integrating factor of this equation is non-zero. Multiplying this equation by its integrating factor converts this into a:

A. Homogeneous differential equation

B. Non-linear differential equation

C. Second order differential equation

D. Exact differential equation

Q6.14.1 MCQ 2014 2M

The integrating factor for the differential equation $\frac{dy}{dx} - \frac{y}{1+x} = (1+x)$ is

A. $\frac{1}{1+x}$ **B.** $(1+x)$ **C.** $x(1+x)$ **D.** $\frac{x}{1+x}$

Q6.14.2 MCQ 2014 2M

The differential equation $\frac{d^2y}{dx^2} + x^2\frac{dy}{dx} + x^3y = e^x$ is a

A. non-linear differential equation of first degree

B. linear differential equation of first degree

C. linear differential equation of second degree

D. non-linear differential equation of second degree

Q6.13.1 MCQ 2013 2M

The solution of the differential equation $\frac{dy}{dx} - y^2 = 0$, given $y = 1$ at $x = 0$ is

A. $\frac{1}{1+x}$ **B.** $\frac{1}{1-x}$ **C.** $\frac{1}{(1-x)^2}$ **D.** $\frac{x^3}{3} + 1$

Q6.13.2 MCQ 2013 2M

The solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} + 0.25y = 0$, given $y = 0$ at $x = 0$ and $\frac{dy}{dx} = 1$ at $x = 0$ is

A. $x\,e^{0.5x} - x\,e^{-0.5x}$ **B.** $0.5x\,e^{x} - 0.5x\,e^{-x}$

C. $x\,e^{0.5x}$ **D.** $-x\,e^{0.5x}$

Q6.12.1 MCQ 2012 1M

If a and b are arbitrary constants, then the solution to the ordinary differential equation $\frac{d^2y}{dx^2} - 4y = 0$ is

A. $y = ax + b$

B. $y = ae^{-x}$

C. $y = a\sin 2x + b\cos 2x$

D. $y = a\cosh 2x + b\sinh 2x$

Q6.11.1 MCQ 2011 2M

Which ONE of the following choices is a solution of the differential equation given below?

$$\frac{dy}{dx} = \frac{y^2}{x} + \frac{y}{x} - \frac{2}{x}$$

Note: c is a real constant

A. $y = \frac{c - x^2}{c + 2x^2}$ **B.** $y = \frac{c + 2x^2}{c - x^2}$

C. $y = \frac{c - x^3}{c + 2x^3}$ **D.** $y = \frac{c + 2x^3}{c - x^3}$

Q6.10.1 MCQ 2010 2M

The solution of the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = 0$$

with the initial conditions $y(0) = 0$, $\frac{dy}{dx}\Big|_{t=0} = -1$, is

A. $-t\sin t$ **B.** $-e^{-t}(1 - \cos t)$

C. $-(t + \sin t)/2$ **D.** $-e^{-t}\sin t$

Q6.09.1 MCQ 2009 2M

The general solution of the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0,$$

with C_1 and C_2 are constants of integration, is

A. $C_1\,e^{-3x} + C_2\,e^{-2x}$ **B.** $C_1\,e^{3x} + C_2\,e^{-2x}$

C. $C_1\,e^{3x} + C_2\,e^{2x}$ **D.** $C_1\,e^{-3x} + C_2\,e^{2x}$

Q6.08.1 MCQ 2008 1M

Which **ONE** of the following is **NOT** an integrating factor for the differential equation $x\,dy - y\,dx = 0$?

A. $\frac{1}{x^2}$ **B.** $\frac{1}{y^2}$ **C.** $\frac{1}{xy}$ **D.** $\frac{1}{(x+y)}$

Q6.08.2 MCQ 2008 1M

Which **ONE** of the following is **NOT** a solution of the differential equation $\frac{d^2y}{dx^2} + y = 1$?

A. $y = 1$ **B.** $y = 1 + \cos x$

C. $y = 1 + \sin x$ **D.** $y = 2 + \sin x + \cos x$

Q6.08.3 MCQ 2008 2M

Which **ONE** of the following transformations $\{u = f(y)\}$ reduces $\frac{dy}{dx} + Ay^3 + By = 0$ to a linear differential equation? (A and B are positive constants)

A. $u = y^{-3}$ **B.** $u = y^{-2}$ **C.** $u = y^{-1}$ **D.** $u = y^2$

Q6.07.1 MCQ 2007 1M

The initial condition for which the following equation

$$(x^2 + 2x)\frac{dy}{dx} = 2(x+1)y;\ y(x_0) = y_0$$

has infinitely many solutions, is

A. $y(x = 0) = 5$ **B.** $y(x = 0) = 1$

C. $y(x = 2) = 1$ **D.** $y(x = -2) = 0$

Q6.07.2 MCQ 2007 2M

The solution of the following differential equation

$$x\frac{dy}{dx} + y(x^2 - 1) = 2x^3$$

is

A. 0 **B.** $2 + ce^{-x^2/2}$

C. $c_1x + c_2x^2$ **D.** $2x + cxe^{-x^2/2}$

Q6.06.1 MCQ 2006 2M

The solution to the following equation is

$$x^2\frac{d^3y}{dx^3} + 2x\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$$

is given by

A. $y = C_1x + C_2x^{-2} + C_3$

B. $y = C_1x^2 + C_2x^{-2} + C_3$

C. $y = C_1x^2 + C_2x^{-1} + C_3$

D. $y = C_1x + C_2x^{-1} + C_3$

Q6.05.1 MCQ 2005 1M

Match the following, where x is the spatial coordinate and t is time

	Group - I		Group - II
P.	Wave equation	1.	$\frac{\partial c}{\partial t} = \alpha\frac{\partial c}{\partial x}$
Q.	Heat equation	2.	$\frac{\partial c}{\partial t} = \alpha^2\frac{\partial^2 c}{\partial x^2}$
		3.	$\frac{\partial^2 c}{\partial t^2} = \alpha^2\frac{\partial c}{\partial x}$
		4.	$\frac{\partial^2 c}{\partial t^2} = \alpha^2\frac{\partial^2 c}{\partial x^2}$

A. P-4, Q-2 **B.** P-2, Q-4

C. P-3, Q-1 **D.** P-1, Q-3

Q6.05.2 MCQ 2005 2M

What condition is to be satisfied so that the solution of the differential equation

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$$

is of the form $y = (C_1 + C_2x)e^{mx}$, where C_1 and C_2 are constants of integration?

A. $a^2 = b$ **B.** $b^2 = a$

C. $a^2 = 4b$ **D.** $b^2 = 4a$

Q6.05.3 MCQ 2005 2M

If $f(x)$ is the solution of the equation

$$\frac{dy}{dx} + 2xy + 2x = 0$$

and $g(x)$ is the solution of the equation

$$\frac{dy}{dx} + 2xy - 2x = 0,$$

and the constant of integration in $f(x)$ is equal to that in $g(x)$, then which of the following is true?

A. $g(x) = f(x) + 2$ **B.** $g(x) = f(x) + 1$

C. $g(x) = f(x) - 1$ **D.** $g(x) = f(x) - 2$

Q6.04.1 MCQ 2004 1M

The differential equation $\frac{d^2y}{dx^2} + \sin x \frac{dy}{dx} + ye^x = \sinh x$

A. first order and linear

B. first order and non-linear

C. second order and linear

D. second order and non-linear

Q6.04.2 MCQ 2004 2M

The differential equation for the variation of the amount of salt X in a tank with time t is given by $\frac{dX}{dt} + \frac{X}{20} = 10$. X is in kg and t is in minutes. Assuming that there is no salt in the tank initially, the time (in min) at which the amount of salt increases to 100 kg is

A. $10 \ln 2$ **B.** $20 \ln 2$

C. $50 \ln 2$ **D.** $100 \ln 2$

Q6.04.3 MCQ 2004 2M

The differential equation $\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2} = 0$ can be reduced to (where α is a constant)

A. $\left(\frac{dy}{dx}\right)^3 = \alpha - \frac{3y^2}{2}$ **B.** $\left(\frac{dy}{dx}\right)^2 = \alpha - 2y$

C. $\left(\frac{dy}{dx}\right) = \frac{\alpha}{y^2}$ **D.** $\left(\frac{dy}{dx}\right) = \frac{\alpha}{y}$

Q6.03.1 MCQ 2003 2M

The value of y as $t \to \infty$ for the following differential equation for an initial value of $y(1) = 0$ is:

$$(4t^2 + 1)\frac{dy}{dx} + 8yt - t = 0$$

A. 1 **B.** $1/2$ **C.** $1/4$ **D.** $1/8$

Q6.03.2 MCQ 2003 1M

The differential equation, $\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$ will have a solution of the form (where C_1 and C_2 are constants).

A. $(C_1 + C_2 t)\, e^{-5t}$ **B.** $C_1\, e^{-2t}$

C. $C_1\, e^{-5t} + C_2\, e^{5t}$ **D.** $C_1\, e^{5t} + C_2\, e^{2t}$

Q6.00.1 MCQ 2000 1M

The integrating factor of differential equation

$$\cos^2 x \frac{dy}{dx} + y = \tan x \text{ is}$$

A. $e^{\tan x}$ **B.** $\cos 2x$

C. $e^{-\tan x}$ **D.** $\sin 2x$

Q6.00.2 MCQ 2000 2M

The general solution of

$$\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0 \text{ is}$$

A. $(c_1 x + c_2)e^x + (c_3 + c_4 x)e^{-x}$

B. $c_1 \cos x + c_2 \sin x$

C. $c_1\, e^{2x} + c_2\, e^{-2x}$

D. $(c_1 + c_2 x)\cos x + (c_3 + c_4 x)\sin x$

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Q6.26.1 NAT 2026 2M

If $\frac{dy}{dt} = 2y$, and the value of y at $t = 0$ is 2, then the value of y at $t = 1$ is _____ (*rounded off to two decimal places*).

Q6.25.1 MCQ 2025 1M

Which one of the following equations is a linear differential equation?

A. $\frac{dy}{dx} + 2x = y^2$ **B.** $x^3\frac{dy}{dx} + xy = x^2$

C. $x\frac{d^2y}{dx^2} + 2y\frac{dy}{dx} = 0$ **D.** $\left(\frac{dy}{dx}\right)^2 + 2x = y$

Q6.25.2 MCQ 2025 2M

The solution of the linear differential equation

$$\frac{dy}{dx} + y = e^x,$$

when $y(0) = 0$, is

A. $y = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$ **B.** $y = \frac{1}{2}e^x + \frac{1}{2}e^{-x}$

C. $y = e^x - e^{-x}$ **D.** $y = e^x + e^{-x}$

Q6.24.1 NAT 2024 2M

The following differential equation governs the evolution of variable $x(t)$ with time t, $t \geq 0$.

$$\frac{d^2x}{dt^2} + 4x = e^{-t}$$

Given the initial conditions $x = 0$ and $\frac{dx}{dt} = 0$ at $t = 0$, the value of x at $t = \pi/8$ is _____. (*Rounded off to 3 decimal places*)

Q6.23.1 MCQ 2023 1M

The non-linear differential equation from the following options is

A. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 10y = 0$

B. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 10y = 0$

C. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 10x = 0$

D. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 10xy = 0$

Q6.22.1 NAT 2022 1M

The order of the following differential equation is _____. [*in integer*]

$$\left(\frac{dy}{dx}\right)^2 + 5\frac{dy}{dx} + 4y = 5x^3$$

Q6.22.2 MCQ 2022 2M

Consider the following ordinary differential equation:

$$4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 0.$$

Given that c_1 and c_2 are constants, the general solution of the differential equation is

A. $y = (c_1 + c_2x)e^x$ **B.** $y = c_1e^{x/2} + c_2e^x$

C. $y = c_1e^x + c_2e^{2x}$ **D.** $y = (c_1 + c_2x)e^{x/2}$

Q6.20.1 MCQ 2020 1M

An integrating factor for the differential equation $\frac{dy}{dx} + my = e^{-mx}$ is

A. e^m **B.** e^{-m} **C.** e^{-mx} **D.** e^{mx}

Q6.20.2 MCQ 2020 2M

General solution of $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$ is

A. $y = \frac{C_1}{x} + \frac{C_2}{x^3}$ **B.** $y = C_1x^2 + \frac{C_2}{x^2}$

C. $y = C_1x + \frac{C_2}{x}$ **D.** $y = C_1x + \frac{C_2}{x^3}$

Q6.19.1 MCQ 2019 1M

If roots of the auxiliary equation of $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$ are real and equal, the general solution of the differential equation is

A. $y = c_1e^{-ax/2} + c_2e^{ax/2}$

B. $y = (c_1 + c_2x)e^{-ax/2}$

C. $y = (c_1 + c_2\ln x)e^{-ax/2}$

D. $y = (c_1\cos x + c_2\sin x)e^{-ax/2}$

Q6.19.2 MCQ 2019 2M

General solution of the Cauchy-Euler equation is $x^2\frac{d^2y}{dx^2} - 7x\frac{dy}{dx} + 16y = 0$ is

A. $y = c_1x^2 + c_2x^4$ **B.** $y = c_1x^2 + c_2x^{-4}$

C. $y = (c_1 + c_2\ln x)x^4$ **D.** $y = c_1x^4 + c_2x^{-4}\ln x$

Q6.18.1 NAT 2018 2M

Consider the differential equation

$$2\frac{d^2y}{dt^2} + 8y = 0$$

with initial conditions: at $t = 0$, $y = 0$ and $\frac{dy}{dt} = 10$. The value of y (up to two decimal places) at $t = 1$ is _____.

Q6.16.1 MCQ 2016 1M

For the two functions

$$f(x, y) = x^3 - 3xy^2 \text{ and } g(x, y) = 3x^2y - y^3$$

which one of the following options is correct?

A. $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial x}$ **B.** $\frac{\partial f}{\partial x} = -\frac{\partial g}{\partial y}$

C. $\frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}$ **D.** $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$

Q6.15.1 MCQ 2015 2M

The solution of $6yy' - 25x = 0$ represents a

A. family of circles **B.** family of ellipses

C. family of parabolas **D.** family of hyperbolas

Q6.13.1 MCQ 2013 1M

The partial differential equation $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2}$ is a

A. linear equation of order 2

B. non-linear equation of order 1

C. linear equation of order 1

D. non-linear equation of order 2

Q6.11.1 MCQ 2011 1M

The solution of the differential equation $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 9x + 6$ with C_1 and C_2 as constants is

A. $y = (C_1x + C_2)e^{-3x}$

B. $y = C_1e^{3x} + C_2e^{-3x} + x$

C. $y = (C_1x + C_2)e^{-3x} + x$

D. $y = (C_1x + C_2)e^{3x} + x$

Q6.10.1 MCQ 2010 1M

Which one of the following differential equations has a solution given by the function $y = 5\sin\left(3x + \frac{\pi}{3}\right)$?

A. $\frac{dy}{dx} - \frac{5}{3}\cos(3x) = 0$ **B.** $\frac{dy}{dx} + \frac{5}{3}\cos(3x) = 0$

C. $\frac{d^2y}{dx^2} + 9y = 0$ **D.** $\frac{d^2y}{dx^2} - 9y = 0$

Q6.10.2 MCQ 2010 2M

The solution of the differential equation

$$\frac{dy}{dx} - y^2 = 1$$

satisfying the condition $y(0) = 1$ is

A. $y = e^{x^2}$ **B.** $y = \sqrt{x}$

C. $y = \cot\left(x + \frac{\pi}{4}\right)$ **D.** $y = \tan\left(x + \frac{\pi}{4}\right)$

Q6.09.1 MCQ 2009 1M

The homogeneous part of the differential equation

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = r$$

(p, q and r are constants) has real and distinct roots if

A. $p^2 - 4q > 0$ **B.** $p^2 - 4q < 0$

C. $p^2 - 4q = 0$ **D.** $p^2 - 4q = r$

Q6.09.2 MCQ 2009 2M

The solution of the differential equation $\frac{d^2y}{dx^2} = 0$ with boundary conditions:

(i) $\frac{dy}{dx} = 1$ at $x = 0$ and

(ii) $\frac{dy}{dx} = 1$ at $x = 1$, is

A. $y = 1$

B. $y = x$

C. $y = x + C$, where C is an arbitrary constant.

D. $y = C_1x + C_2$, where C_1 and C_2 are arbitrary constants.

Q6.08.1 MCQ 2008 2M

For the partial differential equation $\frac{\partial^2 u}{\partial x^2} = \pi^2\frac{\partial u}{\partial t}$ in the domain $0 \le x \le 1$ with boundary conditions $u(0, t) = 0$ and $u(1, t) = 0$ and initial condition $u(x, 0) = \sin(\pi x)$, the solution of the differential equation is

A. $e^{-t}\sin(\pi x)$ **B.** $e^{t}\sin(\pi x)$

C. $e^{-\pi t}\sin(\pi x)$ **D.** $e^{\pi t}\sin(\pi x)$

Q6.08.2 MCQ 2008 2M

The solution of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$ are

A. $e^{-(1+i)x}$, $e^{-(1-i)x}$ **B.** $e^{(1+i)x}$, $e^{(1-i)x}$

C. $e^{-(1+i)x}$, $e^{(1+i)x}$ **D.** $e^{(1+i)x}$, $e^{-(1+i)x}$

Q6.05.1 MCQ 2005 1M

The differential equation

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = C^2\left[\frac{d^2y}{dx^2}\right]^2$$

is of

A. 2nd order and 3rd degree

B. 3rd order and 2nd degree

C. 2nd order and 2nd degree

D. 3rd order and 3rd degree

Q6.94.1 MCQ 1994 1M

Solve for y if $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ with $y(0) = 1$ and $y'(0) = -2$

A. $(1-x)e^{-x}$

B. $(1+x)e^{x}$

C. $(1+x)e^{-x}$

D. $(1-x)e^{x}$

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Answer Key

ELECTRONICS & COMMUNICATION ENGINEERING					
Q6.26.1 A	Q6.25.1 0.25	Q6.24.1 A	Q6.22.1 A,B	Q6.21.1 B	Q6.20.1 B
Q6.20.2 C	Q6.20.3 A	Q6.19.1 C	Q6.19.2 5.24 to 5.26	Q6.18.1 A	Q6.18.2 -0.23 to -0.19
Q6.17.1 D	Q6.17.2 A	Q6.16.1 A	Q6.15.1 B	Q6.15.2 C	Q6.15.3 C
Q6.15.4 0.83 to 0.88	Q6.14.1 A	Q6.14.2 A	Q6.14.3 C	Q6.14.4 B	Q6.14.5 0.53 to 0.55
Q6.13.1 D	Q6.12.1 D	Q6.12.2 D	Q6.11.1 C	Q6.10.1 D	Q6.09.1 B
Q6.09.2 A	Q6.08.1 B	Q6.07.1 D	Q6.06.1 A	Q6.06.2 A	Q6.05.1 B
Q6.05.2 B	Q6.94.1 A				
ELECTRICAL ENGINEERING					
Q6.26.1 A	Q6.26.2 A	Q6.25.1 2	Q6.24.1 B,D	Q6.20.1 0.8774 to 0.8952	Q6.19.1 B
Q6.17.1 A	Q6.16.1 B	Q6.16.2 A	Q6.16.3 7.38	Q6.15.1 -3	Q6.15.2 1.652
Q6.14.1 C	Q6.14.2 C	Q6.11.1 A	Q6.10.1 B	Q6.05.1 A	Q6.05.2 B
INSTRUMENTATION ENGINEERING					
Q6.26.1 C	Q6.26.2 B	Q6.25.1 A	Q6.24.1 33	Q6.23.1 C	Q6.22.1 C
Q6.20.1 A	Q6.18.1 D	Q6.14.1 D	Q6.13.1 A	Q6.13.2 D	Q6.11.1 C
Q6.10.1 C	Q6.08.1 A	Q6.07.1 C	Q6.06.1 A	Q6.05.1 C	
MECHANICAL ENGINEERING					
Q6.26.1 A	Q6.26.2 A	Q6.25.1 B	Q6.25.2 0.45 to 0.55	Q6.24.1 -1.40 to -1.38	Q6.23.1 8.999 to 9.001
Q6.22.1 B	Q6.22.2 C	Q6.21.1 C	Q6.21.2 C	Q6.19.1 C	Q6.19.2 B
Q6.19.3 A	Q6.18.1 C	Q6.18.2 1.47	Q6.17.1 B	Q6.17.2 A	Q6.17.3 93 to 95
Q6.16.1 -1.05 to -0.95	Q6.15.1 C	Q6.15.2 C	Q6.14.1 A	Q6.14.2 34 to 36	Q6.14.3 D

Q6.14.4 A	Q6.14.5 B	Q6.13.1 D	Q6.13.2 B	Q6.12.1 A	Q6.11.1 D
Q6.10.1 B	Q6.09.1 A	Q6.08.1 B	Q6.08.2 C	Q6.08.3 A	Q6.07.1 A
Q6.07.2 C	Q6.06.1 B	Q6.06.2 B	Q6.05.1 D	Q6.05.2 C	Q6.05.3 C
Q6.03.1 A	Q6.01.1 C	Q6.00.1 D	Q6.99.1 C	Q6.98.1 D	Q6.98.2 0.625
Q6.96.1 A	Q6.95.1 A	Q6.95.2 C	Q6.94.1 B	Q6.94.2 A	Q6.94.3 D
Q6.93.1 B	Q6.93.2 B				
CIVIL ENGINEERING					
Q6.26.1 A	Q6.26.2 A,C	Q6.26.3 0.37 to 0.42	Q6.25.1 B	Q6.25.2 5.7 to 5.9	Q6.25.3 3
Q6.25.4 A	Q6.24.1 B	Q6.24.2 A,D	Q6.24.3 B	Q6.23.1 0.446	Q6.23.2 A
Q6.22.1 A	Q6.22.2 A	Q6.21.1 A	Q6.21.2 A	Q6.20.1 B	Q6.20.2 A
Q6.20.3 D	Q6.20.4 A	Q6.20.5 A	Q6.19.1 C	Q6.19.2 5.9 to 6.1	Q6.19.3 D
Q6.18.1 D	Q6.18.2 0.36 to 0.37	Q6.18.3 C	Q6.17.1 35.9 to 36.1	Q6.17.2 D	Q6.17.3 A
Q6.16.1 C	Q6.16.2 B	Q6.16.3 A	Q6.15.1 C	Q6.15.2 18	Q6.14.1 D
Q6.14.2 3.0	Q6.13.1 3.8	Q6.12.1 D	Q6.11.1 D	Q6.10.1 A	Q6.10.2 C
Q6.10.3 C	Q6.09.1 A	Q6.08.1 A	Q6.08.2 D	Q6.08.3 D	Q6.07.1 B
Q6.07.2 D	Q6.06.1 A	Q6.05.1 A	Q6.05.2 A	Q6.04.1 B	Q6.01.1 C
Q6.01.2 C	Q6.00.1 A	Q6.99.1 C	Q6.98.1 A	Q6.97.1 B	Q6.97.2 D
Q6.95.1 C	Q6.95.2 C	Q6.94.1 A	Q6.94.2 C		
CHEMICAL ENGINEERING					
Q6.26.1 2.24 to 2.26	Q6.25.1 B,D	Q6.24.1 4.00 to 4.08	Q6.23.1 -1	Q6.22.1 A	Q6.21.1 0.161 to 0.169
Q6.20.1 162 to 170	Q6.19.1 D	Q6.18.1 C	Q6.17.1 0.49 to 0.51	Q6.16.1 D	Q6.15.1 D
Q6.14.1 A	Q6.14.2 B	Q6.13.1 B	Q6.13.2 C	Q6.12.1 D	Q6.11.1 D

Q6.10.1 D	Q6.09.1 B	Q6.08.1 D	Q6.08.2 D	Q6.08.3 B	Q6.07.1 D
Q6.07.2 D	Q6.06.1 C	Q6.05.1 A	Q6.05.2 C	Q6.05.3 A	Q6.04.1 C
Q6.04.2 B	Q6.04.3 D	Q6.03.1 D	Q6.03.2 A	Q6.00.1 A	Q6.00.2 D
PRODUCTION & INDUSTRIAL ENGINEERING					
Q6.26.1 14.5 to 14.9	Q6.25.1 B	Q6.25.2 A	Q6.24.1 0.064 to 0.065	Q6.23.1 B	Q6.22.1 1
Q6.22.2 D	Q6.20.1 D	Q6.20.2 C	Q6.19.1 B	Q6.19.2 C	Q6.18.1 4.52 to 4.56
Q6.16.1 C	Q6.15.1 D	Q6.13.1 D	Q6.11.1 C	Q6.10.1 C	Q6.10.2 D
Q6.09.1 A	Q6.09.2 C	Q6.08.1 A	Q6.08.2 A	Q6.05.1 C	Q6.94.1 A

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